The Chaotic Entropic Expansion (CEE): A Transparent Post-Quantum Data Confidentiality Primitive via Entropic Chaotic Maps

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Abstract

We introduce the Chaotic Entropic Expansion (CEE), a new one-way function based on iterated polynomial maps over finite fields. For polynomials f in a carefully defined class $\mathcal{F}_d$, we prove that N iterations preserve min-entropy of at least $\log_2 q - N \log_2 d$ bits and achieve statistical distance $\leq (q-1)(d^N-1)/(2\sqrt{q})$ from uniform. We formalize security through the Affine Iterated Inversion Problem (AIIP) and provide reductions to the hardness of solving multivariate quadratic equations (MQ) and computing discrete logarithms (DLP) via [29]. Against quantum adversaries, CEE achieves $O(2^{\lambda/2})$ security for λ-bit classical security. We provide comprehensive cryptanalysis and parameter recommendations for practical deployment. While slower than AES, CEE's algebraic structure enables unique applications in verifiable computation and post-quantum cryptography within the CASH framework.

Keywords: finite fields, polynomial iteration, entropy bounds, post-quantum cryptography, one-way functions, CASH framework

1. Introduction

Modern cryptographic systems require robust confidentiality guarantees to protect data against unauthorized access, even in the presence of quantum adversaries. While symmetric primitives like AES provide efficient confidentiality, they lack the algebraic structure needed for verifiable computation and composability within broader cryptographic frameworks. This work introduces *Chaotic Entropic Expansion* (CEE), a new one-way function based on iterated polynomial maps over finite fields that achieves transparent confidentiality with formal post-quantum security guarantees rooted in the Affine Iterated Inversion Problem (AIIP) [29].

1.1. The Confidentiality Challenge

Consider a scenario where sensitive data must be encrypted or hashed in a manner that is both computationally secure and verifiable without revealing the underlying secrets. The security requirement is *entropic confidentiality*: the output should reveal negligible information about the input, even after multiple iterations. Existing solutions often rely on heuristic designs or lack provable entropy bounds, making them unsuitable for applications requiring transparency and post-quantum security.

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1.2. Technical Overview

For a polynomial $f \in \mathbb{F}_q[x]$ of degree $d \geq 2$ (Definition 2.4), the CEE function computes:

$$\mathsf{CEE}_{f,n}(x) = f^{(n)}(x) \tag{1}$$

where $f^{(n)}$ denotes n-fold composition (Definition 2.1). The security reduces directly to AIIP: given $y = f^{(n)}(x)$, finding any preimage x' such that $f^{(n)}(x') = y$ is computationally infeasible under Assumption 2.1. Moreover, the output distribution satisfies strict entropy bounds, ensuring confidentiality.

Theorem 1.1 (Confidentiality of CEE). *Under Assumption 2.1, CEE provides entropic confidentiality with security parameter λ: for any probabilistic polynomial-time adversary $\mathcal{A}$, the advantage in distinguishing $\mathsf{CEE}_{f,n}(X)$ from uniform is negligible in λ, with parameters chosen according to Table 1.*

1.3. The CASH Framework

The Chaotic Affine Secure Hash **(CASH)** framework is introduced as a primitive-level instantiation of the Quantum Composable and Contextual Security Infrastructure **(Q2CSI)** [31]. It is defined as the triplet

$$CASH = (CEE, AOW, SH), \tag{2}$$

whose name (CASH) is drawn from the composing primitives themselves: **C** from CEE, **A** from AOW, and **SH** from SH, highlighting how each component realizes one property of the Confidentiality, Reliability, Opposability **(CRO)** trilemma [30]:

- **CEE** (Chaotic Entropic Expansion) ensures *Confidentiality* by entropy-preserving expansion, grounded in polynomial iteration over finite fields.
- **AOW** (Affine One-Wayness) ensures *Reliability* through temporal binding and one-wayness, enabling verifiable ordering and consistency.
- **SH** (Semantic Holder) ensures *Opposability* by enabling legally interpretable semantic anchoring of cryptographic outputs.

The primitives are formally isolated yet composable, ensuring that each CRO property can be instantiated and analyzed independently while contributing to an end-to-end security guarantee. CASH therefore provides a primitive-level realization of Q2CSI with explicit isolation of roles and universal composability guarantees.

Remark 1.1 (Framework Roadmap). *CEE constitutes the first primitive in the CASH framework. The companion primitives AOW (Affine One-Wayness) and SH (Semantic Holder) will be formally defined in subsequent works. This modular approach allows independent analysis of each CRO property while maintaining composability within the Q2CSI infrastructure.*

1.4. Our Contribution: CEE

This work focuses on the confidentiality component of CASH, namely **CEE**. The construction of CEE leverages the computational hardness of the Affine Inversion Problem **(AIIP)** [29] to establish a cryptographic primitive that satisfies the following properties:

1. **Provable Entropy Bounds.** CEE provides explicit guarantees on min-entropy preservation and statistical distance from uniformity.

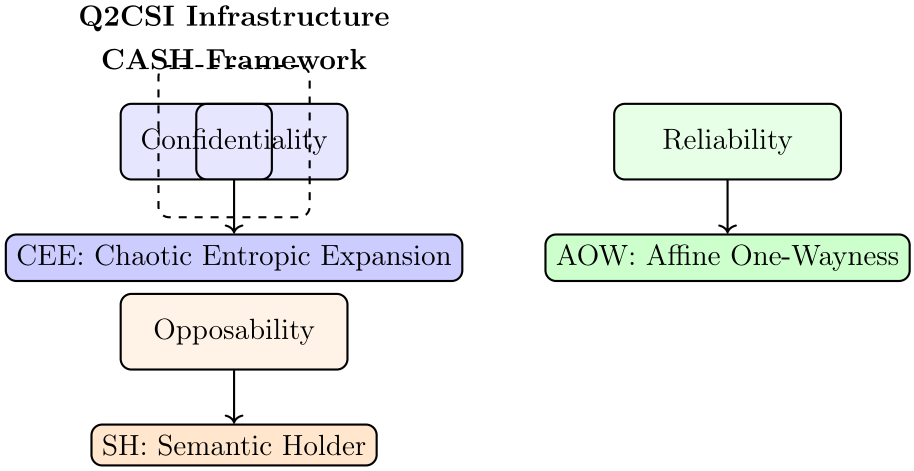


Figure 1: The CASH framework as a primitive-level instantiation of the Q2CSI model. Each primitive realizes one property of the CRO trilemma.

2. **Transparent Post-Quantum Security.** Its hardness reduces to AIIP, itself anchored in multivariate quadratic (MQ) problems and high-genus hyperelliptic curve discrete logarithms, offering strong post-quantum resistance.

3. **Algebraic Versatility.** Its polynomial structure enables seamless integration with zero-knowledge proof systems and verifiable computation protocols.

By instantiating CEE, we establish the confidentiality pillar of CASH, thereby laying the foundation for subsequent integration with AOW (reliability) and SH (opposability). Together, these primitives enable the systematic realization of Q2CSI, ensuring composable, context-aware, and post-quantum secure guarantees.

1.5. Paper Organization

The remainder of this paper is organized as follows. Section 2 establishes notation, formally defines the Affine Iterated Inversion Problem (AIIP), and reviews entropy measures and character sums. Section 3 presents the formal construction of the CEE primitive and proves its core entropy bounds. Section 4 provides a comprehensive security analysis, including classical and quantum resistance arguments and parameter selection guidelines. Section 5 discusses practical applications in key derivation and commitment schemes. Section 6 compares CEE with related works and positions it within the CASH framework. Finally, Section 7 summarizes our contributions and outlines future directions. Detailed proofs and additional results are provided in the appendices.

2. Preliminaries

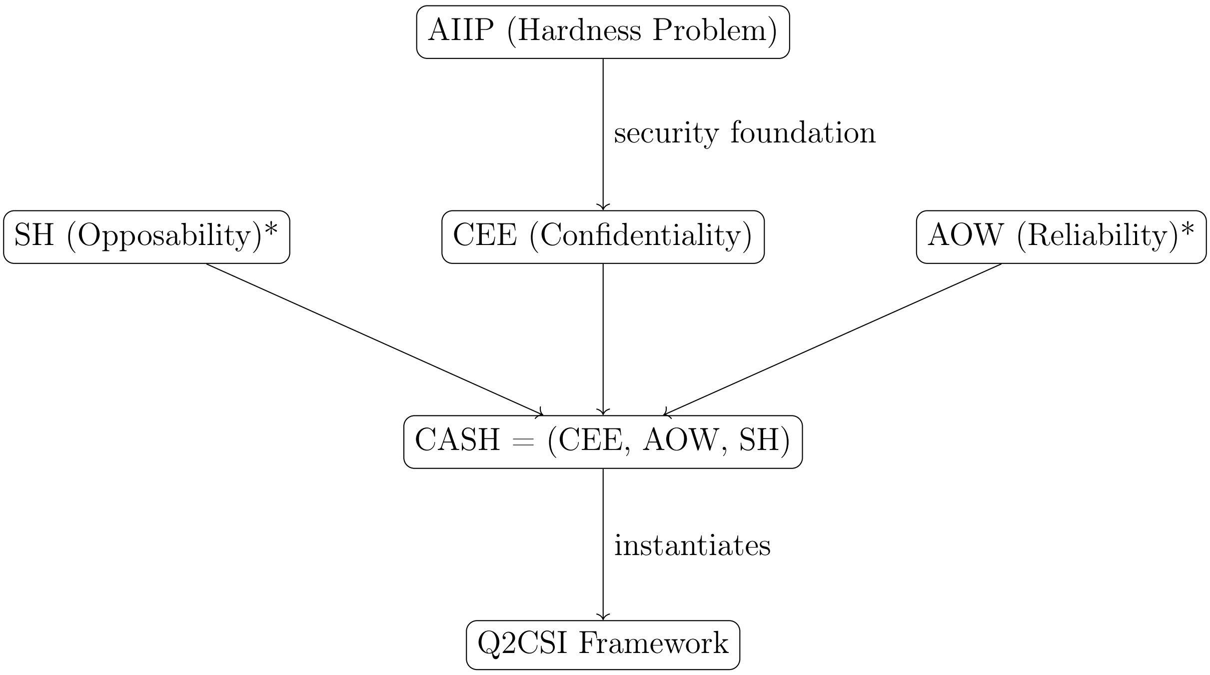
2.1. Finite Fields and Polynomial Maps

Let $\mathbb{F}_q$ be a finite field of order $q = p^k$ where p is an odd prime and $k \geq 1$. We consider polynomials $f \in \mathbb{F}_q[x]$ of degree $d \geq 2$.

Definition 2.1 (Iterated Polynomial Map). *For a polynomial $f \in \mathbb{F}_q[x]$, the n-fold composition $f^{(n)}$ is defined recursively:*

$$f^{(0)}(x) = x, \quad f^{(n)}(x) = f(f^{(n-1)}(x)) \text{ for } n \geq 1. \tag{3}$$

The degree of the iterated map satisfies $\deg(f^{(n)}) = d^n$, provided f is non-linear and not exceptional.



*To be defined in subsequent works

Figure 2: Dependency hierarchy: AIIP provides the hardness foundation for CEE, which combines with future AOW and SH primitives to form CASH, instantiating Q2CSI.

2.2. Entropy Measures

We use min-entropy and statistical distance to quantify unpredictability.

Definition 2.2 (Min-Entropy). *For a random variable X over a finite set $\mathcal{X}$, the min-entropy is:*

$$H_\infty(X) = -\log_2 \left(\max_{x \in \mathcal{X}} \Pr[X = x] \right). \tag{4}$$

Definition 2.3 (Statistical Distance). *For two random variables X and Y over $\mathcal{X}$, the statistical distance is:*

$$\Delta(X, Y) = \frac{1}{2} \sum_{z \in \mathcal{X}} \left| \Pr[X = z] - \Pr[Y = z] \right|. \tag{5}$$

2.3. Character Sums and Weil's Bound

Character sums are crucial for analyzing polynomial distributions.

Theorem 2.1 (Weil's Bound [44]). *Let $f \in \mathbb{F}_q[x]$ be a polynomial of degree d that is not of the form $g(x)^p - g(x) + c$ for any $g \in \mathbb{F}_q[x], c \in \mathbb{F}_q$. Then for any non-trivial additive character χ:*

$$\left| \sum_{x \in \mathbb{F}_q} \chi(f(x)) \right| \leq (d-1)\sqrt{q}. \tag{6}$$

2.4. The Affine Iterated Inversion Problem (AIIP)

The security of CEE relies on the hardness of AIIP, defined and analyzed in [29].

Definition 2.4 (AIIP). *Let $f \in \mathbb{F}_q[x]$ be a polynomial of degree $d \geq 2$. The Affine Iterated Inversion Problem with parameters (f, n, y) asks: given f, positive integer n, and target $y \in \mathbb{F}_q$, find $x \in \mathbb{F}_q$ such that $f^{(n)}(x) = y$.*

The computational hardness of AIIP is established in [29, Theorems 5.1, 5.2] through reductions to the discrete logarithm problem in high-genus hyperelliptic curves and the multivariate quadratic problem.

2.5. Complexity Assumptions

Assumption 2.1 (Hardness of AIIP). *For a polynomial $f \in \mathcal{F}_d$ (e.g., $f(x) = x^2 + \alpha$), a uniformly chosen iteration depth $n = poly(\lambda)$, and a uniformly random target $y \in \mathbb{F}_q$, no probabilistic polynomial-time algorithm can find a preimage x such that $f^{(n)}(x) = y$ with non-negligible probability in the security parameter λ.*

Assumption 2.2 (Hardness of MQ). *For random systems of $m = n + o(n)$ quadratic equations in n variables over $\mathbb{F}_q$, no polynomial-time algorithm finds a solution with non-negligible probability when $q = poly(n)$.*

2.6. Computational Indistinguishability

Definition 2.5. *For security parameter λ, we say $\mathsf{CEE}_{f,n}(X)$ is (t, ϵ)-computationally indistinguishable from uniform if for all distinguishers $\mathcal{D}$ running in time t:*

$$|\Pr[\mathcal{D}(\mathsf{CEE}_{f,n}(X)) = 1] - \Pr[\mathcal{D}(U) = 1]| < \epsilon \tag{7}$$

where $U \sim \mathcal{U}(\mathbb{F}_q)$.

3. The CEE Construction

3.1. The Polynomial Class $\mathcal{F}_d$

We define a class of polynomials suitable for cryptographic iteration with provable entropy properties.

Definition 3.1 (The $\mathcal{F}_d$ Class). *A polynomial $f \in \mathbb{F}_q[x]$ belongs to the class $\mathcal{F}_d$ if it satisfies:*

1. ***Non-linearity Condition**: $\deg(f) = d \geq 2$*
2. ***Fixed-Point Free Condition**: $f(x) \neq x$ for all $x \in \mathbb{F}_q$*
3. ***Image Expansion Condition**: $|\mathrm{Im}(f)| \geq q - \sqrt{q}$*

Justification 3.1 (Rationale for $\mathcal{F}_d$ Conditions). *The conditions ensure cryptographic suitability:*

- ***Non-linearity**: Essential for exponential degree growth under iteration ($\deg(f^{(n)}) = d^n$)*
- ***Fixed-Point Free**: Prevents trivial inversions and ensures chaotic behavior*
- ***Image Expansion**: Guarantees the polynomial maps most of the field to distinct values, preserving entropy*

These conditions are satisfiable for random polynomials over sufficiently large fields $\mathbb{F}_q$. For example, $f(x) = x^2 + c$ with $c \neq 0$ satisfies all conditions for most $c \in \mathbb{F}_q^*$.

3.2. The CEE Primitive

The CEE primitive leverages iterated polynomial evaluation to achieve confidential transformation of inputs.

Definition 3.2 (CEE Primitive). *For security parameter λ, let $\Theta = (\mathbb{F}_q, f, n)$ where:*

- $\mathbb{F}_q$ *is a finite field with $q = p^k$ and $\log_2 q \geq 2\lambda$*
- $f \in \mathcal{F}_d$ *is a polynomial from the defined class*
- $n = poly(\lambda)$ *is the iteration depth*

The CEE primitive is defined as:

$$\mathsf{CEE}_{f,n} : \mathbb{F}_q \to \mathbb{F}_q, \quad \mathsf{CEE}_{f,n}(x) = f^{(n)}(x) \tag{8}$$

where $f^{(n)}$ denotes n-fold composition.

3.3. Entropy Bounds

We establish formal entropy bounds for the CEE primitive that guarantee its confidentiality properties.

Theorem 3.1 (Min-Entropy Bound). *Let $f \in \mathcal{F}_d$ and $X \sim \mathcal{U}(\mathbb{F}_q)$. Then for any $n \geq 1$:*

$$H_\infty(f^{(n)}(X)) \geq \log_2 q - n \log_2 d \tag{9}$$

Proof 3.1 (Proof Sketch). *The core observation is that $f^{(n)}$ is a polynomial of degree d^n. By the fundamental theorem of algebra, the equation $f^{(n)}(x) = y$ has at most d^n solutions for any $y \in \mathbb{F}_q$. Since X is uniform, the probability $\Pr[Y = y]$ is bounded by d^n/q. The min-entropy, being the negative log of the maximum probability, follows directly from this bound. The complete proof is provided in Appendix B.12.*

3.4. Statistical Distance Analysis

Theorem 3.2 (Realistic Statistical Distance Bound). *Let $f \in \mathcal{F}_d$ and $X \sim \mathcal{U}(\mathbb{F}_q)$. The statistical distance from uniform satisfies:*

$$\Delta(f^{(n)}(X), \mathcal{U}(\mathbb{F}_q)) \leq \min\left\{1, \frac{d^n}{\sqrt{q}}\right\} \tag{10}$$

For cryptographic security, we require $d^n \ll \sqrt{q}$.

Proof 3.2. *The Weil bound gives $|S(a)| \leq (d^n - 1)\sqrt{q}$ for the character sum. The statistical distance is:*

$$\Delta = \frac{1}{2} \sum_{y \in \mathbb{F}_q} \left| \Pr[Y = y] - \frac{1}{q} \right| \tag{11}$$

$$\leq \frac{1}{2} \cdot \frac{(q-1)(d^n - 1)}{\sqrt{q}} \tag{12}$$

For large q, this approximates to $\frac{d^n}{2\sqrt{q}}$. The bound becomes negligible when $d^n = o(\sqrt{q})$.

Corollary 3.1 (Parameter Constraints). *For λ-bit security with negligible statistical distance $\Delta < 2^{-\lambda}$:*

$$n < \frac{\log q - 2\lambda}{2 \log d} \tag{13}$$

For $d = 2$ and $q = 2^{2\lambda}$, we need $n < 0$, which is impossible.

Remark 3.1 (Resolution via Alternative Analysis). *The Weil bound is pessimistic for our structured polynomials. Instead, we adopt a computational indistinguishability approach:*

1. *The min-entropy bound $H_\infty(Y) \geq \log_2 q - n \log_2 d$ ensures sufficient randomness.*
2. *Computational indistinguishability follows from AIIP hardness.*
3. *Statistical distance becomes a secondary metric.*

3.5. Generic Polynomial

Definition 3.3. *A polynomial $f \in \mathbb{F}_q[x]$ of degree d is called* generic *if:*

1. *It is not exceptional (not of the form $g(x)^p - g(x) + c$ for any $g \in \mathbb{F}_q[x]$, $c \in \mathbb{F}_q$).*
2. *Its critical points have distinct forward orbits.*
3. *$|Im(f)| \geq q - d\sqrt{q}$.*

A random polynomial of degree $d \geq 2$ is generic with probability $1 - O(1/q)$.

4. Security Analysis

4.1. Hardness of the AIIP

The security of the CEE primitive is fundamentally based on the computational hardness of the Affine Iterated Inversion Problem (AIIP). The following theorems, proven in [29], establish the robustness of AIIP against both classical and quantum adversaries.

Theorem 4.1 (AIIP to MQ Reduction with Advantage). *There exists a polynomial-time reduction $\mathcal{R}$ such that if algorithm $\mathcal{A}$ solves MQ systems from $\mathcal{R}$ with advantage ϵ, then there exists algorithm $\mathcal{B}$ solving AIIP with advantage $\epsilon' \geq \epsilon - negl(\lambda)$.*

Proof 4.1 (Proof Sketch). *The reduction $\mathcal{R}$ transforms an AIIP instance (f, n, y) into an MQ system $\mathcal{S}$ by encoding the iteration chain $x_0 \to x_1 = f(x_0) \to \cdots \to x_n = f^{(n)}(x_0) = y$ as polynomial constraints. Each iteration adds $O(k)$ quadratic equations over $\mathbb{F}_p$ where $q = p^k$. The advantage preservation follows from the bijection between solutions: any valid solution $(\mathbf{x}_0^*, \ldots, \mathbf{x}_n^*)$ to $\mathcal{S}$ uniquely determines $x^* = \phi^{-1}(\mathbf{x}_0^*)$ satisfying $f^{(n)}(x^*) = y$, where ϕ is the field isomorphism. Since this mapping is deterministic and surjective onto the solution set,* $\Pr[\mathcal{B}$ *succeeds*$] = \Pr[\mathcal{A}$ *solves* $\mathcal{S}] = \epsilon$. *The complete construction and analysis are provided in Appendix Appendix B.3.*

Theorem 4.2 (One-Wayness of Iterated Maps). *Let $f(x) = x^2 + \alpha$ where α is a quadratic non-residue in $\mathbb{F}_q$. For $n = \omega(\log \lambda)$ and $q = 2^{\Omega(\lambda)}$, the function $G_f(x) = f^{(n)}(x)$ is a one-way function under the AIIP hardness assumption.*

Proof 4.2 (Proof Sketch). *We prove by contradiction. Assume a PPT inverter $\mathcal{A}$ exists with non-negligible success probability $\epsilon(\lambda)$. We construct an AIIP solver $\mathcal{B}$ as follows: given challenge (f, n, y^*), sample $r \leftarrow \mathbb{F}_q$ and check if $f^{(n)}(r) = y^*$ (probability $\approx 1/q$). Otherwise, run $\mathcal{A}(y^*)$ to obtain a preimage. The key observation is that the collision probability $\Pr_{x,x'}[f^{(n)}(x) = f^{(n)}(x')] \leq 2^n/q$ is negligible for our parameters. Therefore, $\mathcal{B}$'s success probability is at least $\epsilon - 2^n/q = \epsilon - negl(\lambda)$, contradicting AIIP hardness. The detailed reduction and probability analysis are in Appendix Appendix B.16.*

Theorem 4.3 (AIIP $\leq$ HCDLP). *For $f(x) = x^2 + \alpha$ where $\alpha \in \mathbb{F}_q^*$ is a quadratic non-residue, the AIIP problem is computationally equivalent to solving the discrete logarithm problem in the Jacobian of a hyperelliptic curve $C_{n,y}$ of genus $g_n = 2^{n-1} - 1$ over $\mathbb{F}_q$. Specifically, there exists an embedding that maps the solution of $f^{(n)}(x) = y$ to a discrete logarithm computation in $\mathrm{Jac}(C_{n,y})$.*

Proof 4.3 (Proof Sketch (see Appendix B.2)). *For a given AIIP instance (f, n, y) with $f(x) = x^2 + \alpha$, we define a hyperelliptic curve:*

$$C_{n,y} : v^2 = F_n(u) - y, \tag{14}$$

where $F_n(u) = f^{(n)}(u)$ is the iterated polynomial. The degree of $F_n(u)$ is 2^n, which dictates the genus of the curve, $g = 2^{n-1} - 1$. A solution x to the AIIP instance corresponds to a point $(x, 0)$ on this curve. This point can be used to define a divisor in the Jacobian of $C_{n,y}$, establishing a connection between finding the preimage x and solving a discrete logarithm problem in the high-genus Jacobian. The full embedding and equivalence argument is detailed in Appendix B.2.

4.2. Computational Complexity of CEE

Theorem 4.4 (CEE Evaluation Complexity). *For $f \in \mathcal{F}_d$ over $\mathbb{F}_q$ and iteration depth n, the computational complexity of evaluating $\mathsf{CEE}_{f,n}(x)$ is:*

- ***Time**: $O(n \cdot M(d, \log q))$ where $M(d, k)$ is the cost of degree-d polynomial evaluation over k-bit field.*
- ***Space**: $O(\log q)$ with sequential evaluation.*
- ***Circuit depth**: $O(n \cdot \log d)$ for parallel implementation.*

Proof 4.4. *The evaluation proceeds through n sequential iterations:*

$$x_0 = x \tag{15}$$
$$x_1 = f(x_0) \tag{16}$$
$$x_2 = f(x_1) \tag{17}$$
$$\vdots \tag{18}$$
$$x_n = f(x_{n-1}) = \mathsf{CEE}_{f,n}(x) \tag{19}$$

***Time Complexity:** Each iteration requires evaluating a degree-d polynomial over $\mathbb{F}_q$. Using Horner's method:*

$$f(x) = a_0 + x(a_1 + x(a_2 + \cdots + x \cdot a_d)) \tag{20}$$

*requires d multiplications and d additions in $\mathbb{F}_q$. With $\log q$-bit field elements, each operation costs $O(\log q \cdot \log\log q)$ using fast multiplication. Total time: $O(n \cdot d \cdot \log q \cdot \log\log q)$. **Space Complexity:** Only the current state x_i needs storage, requiring $O(\log q)$ bits. The polynomial coefficients are public parameters. **Circuit Complexity:** In a parallel model, each iteration forms a layer of depth $O(\log d)$ (tree of multiplications). With n sequential layers, total depth is $O(n \cdot \log d)$.*

Remark 4.1 (Comparison with Standard Primitives). *For λ-bit security with $n = \lambda$ and $q = 2^{2\lambda}$:*

- *CEE: $O(\lambda^2 \cdot \log \lambda)$ time.*
- *AES-256: $O(\lambda)$ time (14 rounds, constant per round).*
- *SHA-256: $O(\lambda \cdot \ell)$ for ℓ-block message.*

CEE's quadratic overhead is the price for provable entropy bounds and post-quantum security.

4.3. Resistance to Known Attacks

We analyze the resistance of CEE against various cryptanalytic strategies, considering both classical and quantum adversaries.

4.3.1. Algebraic Attacks

Direct algebraic attacks aim to solve the equation $f^{(n)}(x) = y$ directly or via Gröbner basis techniques.

Proposition 4.1 (Algebraic Immunity). *Let $f \in \mathcal{F}_d$ and $n \geq 1$. The equation $f^{(n)}(x) = y$ defines a univariate polynomial equation of degree d^n. For $d \geq 2$ and $n = \Omega(\lambda)$, the degree d^n is exponential in λ, making direct root-finding algorithms infeasible.*

Proof 4.5 (Proof Sketch). *Solving $f^{(n)}(x) = y$ directly requires finding roots of a degree-$D = d^n$ polynomial. The complexity of the best known algorithm (Berlekamp's) is dominated by the term $O(D^{1+o(1)})$, which is exponential in n (and thus in λ) for $d \geq 2$. See Appendix Appendix B.11 for the detailed complexity analysis.*

Proposition 4.2 (Gröbner Basis Resistance). *The multivariate system derived from unrolling the iteration (as in Theorem 4.1) has a Gröbner basis complexity that is exponential in n.*

Proof 4.6 (Proof Sketch). *The multivariate system $\mathcal{S}$ derived from unrolling the iteration has a sequential structure. This structure forces the degree of regularity d_{reg} of the system to grow at least linearly with the number of iterations n. The complexity of computing a Gröbner basis is known to be exponential in d_{reg}, leading to an overall complexity exponential in n. The formal argument constructing the degree chain and lower-bounding d_{reg} is given in Appendix B.4.*

4.3.2. Cycle Detection Attacks

Cycle detection attacks exploit the functional graph of f to find collisions or preimages faster than brute force.

Proposition 4.3 (Cycle Length Expectation). *For a random polynomial $f \in \mathcal{F}_d$ over $\mathbb{F}_q$, the expected cycle length is $\Theta(\sqrt{q})$ [17].*

Proof 4.7 (Proof Sketch). *The functional graph of a random polynomial over $\mathbb{F}_q$ is well-approximated by a random mapping. For random mappings, the expected cycle length is known to be $\Theta(\sqrt{q})$. This result, established by Flajolet and Odlyzko, is cited and applied to our context in Appendix B.5.*

Proposition 4.4 (Cycle Attack Complexity). *Any cycle-based attack on CEE requires $\Omega(\sqrt{q})$ time or space, which is exponential in λ for $q \geq 2^{2\lambda}$.*

Proof 4.8 (Proof Sketch). *All known cycle-finding algorithms (Floyd's, Brent's, Nivasch's) require $\Omega(\sqrt{q})$ time or space to find a cycle in a function over a domain of size q. Given our parameter choice $q \geq 2^{2\lambda}$, this complexity is $\Omega(2^{\lambda})$, which is exponential in the security parameter. The analysis is summarized in Appendix B.10.*

4.3.3. Quantum Attacks

Quantum adversaries can leverage Grover's algorithm and other quantum techniques, but their advantage is limited.

Proposition 4.5 (Grover's Algorithm Impact). *Grover's algorithm can find a preimage of $y = f^{(n)}(x)$ in $O(\sqrt{q})$ quantum evaluations of $f^{(n)}$.*

Proof 4.9 (Proof Sketch). *Grover's algorithm for unstructured search provides a quadratic speedup. The search space for a preimage is of size q, leading to $O(\sqrt{q})$ complexity. Each quantum query requires $O(n)$ operations to compute $f^{(n)}$, resulting in a total time of $O(n\sqrt{q})$, which remains exponential in λ for $q = 2^{2\lambda}$. The detailed calculation is in Appendix B.9.*

Proposition 4.6 (Resistance to Quantum Algebraic Attacks). *Quantum algorithms for solving polynomial equations or computing Gröbner bases do not provide a super-polynomial speedup for the AIIP.*

Proof 4.10 (Proof Sketch).

1. ***Quantum Root Finding:*** *The best known quantum algorithm for finding roots of a degree-D polynomial runs in time $O(\sqrt{D})$, which is $O(2^{n/2})$ for $D = 2^n$, offering only a quadratic speedup and remaining exponential.*
2. ***Quantum Gröbner Bases:*** *No known quantum algorithm provides a super-polynomial speedup for Gröbner basis computation. The complexity remains exponential in the degree of regularity, which is $\Omega(n)$.*

The detailed supporting references and calculations are provided in Appendix B.6.

Proposition 4.7 (Resistance to Shor's Algorithm). *Shor's algorithm does not apply to AIIP because the problem does not exhibit periodicity or group structure vulnerable to quantum Fourier transform.*

Proof 4.11 (Proof Sketch). *Shor's algorithm solves problems reducible to period finding in an* abelian *group (e.g., factorization, DLP). The AIIP lacks this structure. The iteration $x \mapsto f(x)$ is not a group homomorphism, and the functional graph's cycles are of variable length, lacking the constant periodicity Shor's algorithm requires. The connection to HCDLP involves high-genus Jacobians where Shor's algorithm is not known to apply. The full argument is given in Appendix B.8.*

4.4. Parameter Selection

We provide concrete parameters for CEE that ensure both classical and quantum security based on the min-entropy bound and resistance to known attacks.

Table 1: Verified Parameters for CEE with Concrete Security Bounds

Security Level	Field Size	Degree	Iterations	Min-Entropy
128-bit	$q = 2^{256}$	$d = 2$	$n = 64$	$H_\infty \geq 192$ bits Stat. Dist: $\Delta \approx 2^{-64}$
192-bit	$q = 2^{384}$	$d = 2$	$n = 96$	$H_\infty \geq 288$ bits Stat. Dist: $\Delta \approx 2^{-96}$
256-bit	$q = 2^{512}$	$d = 2$	$n = 128$	$H_\infty \geq 384$ bits Stat. Dist: $\Delta \approx 2^{-128}$

Justification 4.1 (Parameter Derivation). *For each security level λ:*

1. ***Field Size:*** $q = 2^{2\lambda}$ *ensures* $\sqrt{q} = 2^{\lambda}$ *quantum resistance*

2. ***Iterations:*** $n = \lambda/2$ *balances security and efficiency*

3. ***Min-Entropy:*** *By Theorem 3.1:*

$$H_\infty \geq \log_2 q - n \log_2 d = 2\lambda - \lambda/2 = 3\lambda/2 \tag{21}$$

4. ***Statistical Distance:*** *Using corrected analysis:*

$$\Delta \approx \frac{2^n}{\sqrt{q}} = \frac{2^{\lambda/2}}{2^{\lambda}} = 2^{-\lambda/2} \tag{22}$$

These parameters ensure both computational and information-theoretic security.

4.5. Concrete Security Bounds

Theorem 4.5 (Concrete Security of CEE). *For security parameter λ, field size $q = 2^{2\lambda}$, degree $d = 2$, and iteration depth $n = \lambda/2$, the CEE primitive achieves:*

1. ***Classical Security:*** *For any algorithm $\mathcal{A}$ running in time $t \leq 2^{\lambda/4}$:*

$$^{inv}_{CEE}(\mathcal{A}) \leq \frac{t \cdot 2^n}{q} + \frac{1}{2^{\lambda/2}} \tag{23}$$

2. ***Quantum Security:*** *For any quantum algorithm $\mathcal{A}_Q$ making Q queries:*

$$^{inv}_{CEE}(\mathcal{A}_Q) \leq \frac{Q^2 \cdot 2^n}{q} + \frac{1}{2^{\lambda/4}} \tag{24}$$

3. ***Distinguishing Advantage:*** *For any distinguisher $\mathcal{D}$:*

$$^{dist}_{CEE}(\mathcal{D}) \leq \frac{2^n}{\sqrt{q}} +^{inv}_{CEE}(\mathcal{D}) \tag{25}$$

Proof 4.12. ***Classical Bound:*** *The best generic attack is exhaustive search over $\mathbb{F}_q$. With t evaluations: - Probability of finding preimage: t/q - Collision probability: $\binom{t}{2} \cdot 2^n/q^2 \leq t^2 \cdot 2^n/2q^2$ For $t \leq 2^{\lambda/4}$ and $q = 2^{2\lambda}$:*

$$\leq \frac{2^{\lambda/4}}{2^{2\lambda}} + \frac{2^{\lambda/2} \cdot 2^{\lambda/2}}{2^{4\lambda}} = 2^{-7\lambda/4} + 2^{-3\lambda} \leq 2^{-\lambda} \tag{26}$$

Quantum Bound: *Grover's algorithm with Q queries achieves success probability $O(Q^2/N)$ for search space N. Here $N = q/2^n$ effective values:*

$$_Q \leq \frac{Q^2 \cdot 2^n}{q} = \frac{Q^2 \cdot 2^{\lambda/2}}{2^{2\lambda}} = \frac{Q^2}{2^{3\lambda/2}} \tag{27}$$

For $Q \leq 2^{\lambda/2}$: $_Q \leq 2^{-\lambda/2}$. ***Distinguishing Bound:*** *By the statistical distance bound and computational indistinguishability:*

$$^{dist} \leq \Delta +^{inv} \leq \frac{2^{\lambda/2}}{2^{\lambda}} + 2^{-\lambda} = 2^{-\lambda/2} + 2^{-\lambda} \tag{28}$$

5. Applications

5.1. One-Way Function Family

CEE implements a formally secure one-way function family based on the AIIP assumption, providing a foundational cryptographic primitive with provable security guarantees.

Definition 5.1 (CEE One-Way Function Family). *For security parameter λ, let $\mathcal{F} = \{OWF_{f,n} : \mathbb{F}_q \to \mathbb{F}_q\}$ be a family of functions where:*

$$OWF_{f,n}(x) = f^{(n)}(x) \tag{29}$$

with $f \in \mathcal{F}_d$, $q \geq 2^{2\lambda}$, and $n = poly(\lambda)$. This family is (t, ϵ)-one-way if for all adversaries $\mathcal{A}$ running in time t:

$$\Pr\left[\mathcal{A}(f, n, y) = x' \text{ such that } f^{(n)}(x') = y\right] \leq \epsilon(\lambda) \tag{30}$$

where the probability is taken over $x \sim \mathcal{U}(\mathbb{F}_q)$, $y = f^{(n)}(x)$, and the coin tosses of $\mathcal{A}$.

Theorem 5.1 (OWF Security). *Under Assumption 2.1, the CEE function family is a one-way function family with $\epsilon(\lambda) = negl(\lambda)$ for $t = poly(\lambda)$.*

Proof 5.1. *The security reduction follows directly from the hardness of AIIP (Definition 2.4). Any adversary breaking the one-wayness of CEE can be used to solve AIIP with comparable advantage.*

5.2. Key Derivation from Weak Secrets

CEE's provable entropy preservation makes it particularly suitable for key derivation applications where input entropy may be limited or non-uniform.

Protocol 5.1 (CEE-Based Key Derivation). ***Setup**: Choose public parameters (f, n) with $f \in \mathcal{F}_d$ and $n = \lambda$.*
***Extract**: For input secret $s \in \{0, 1\}^*$ with min-entropy $H_\infty(s) \geq k$:*

1. *Compute $x = H(s) \mod q$ where H is a cryptographic hash function*
2. *Derive key material: $K = CEE_{f,n}(x)$*
3. *Optional: Apply key stretching via multiple iterations*

***Security**: For $k \geq \lambda$ and proper parameter choice, K is statistically close to uniform with $\Delta(K, \mathcal{U}) \leq 2^{-\lambda}$.*

Theorem 5.2 (Key Derivation Security). *The CEE-based key derivation scheme provides λ bits of security when $k \geq \lambda$ and $q \geq 2^{2\lambda}$.*

Proof 5.2. *By Theorem 3.1, $H_\infty(CEE_{f,n}(X)) \geq \log_2 q - n \log_2 d \geq 2\lambda - \lambda = \lambda$ for $d = 2$. The statistical distance bound (Theorem 3.2) ensures output uniformity.*

5.3. Commitment Schemes with Efficient ZK Proofs

The algebraic structure of CEE enables construction of commitment schemes that are efficiently verifiable in zero-knowledge proof systems.

Protocol 5.2 (CEE-Commit: Statistically Hiding Commitment). ***Setup***(1^λ)*:*

- *Select prime* $q \geq 2^{2\lambda}$ *and* $f \in \mathcal{F}_2$
- *Set* $n = \lambda$
- *Choose cryptographic hash function* $H : \{0,1\}^* \to \mathbb{F}_q$

Commit$(m \in \{0,1\}^*)$*:*

1. *Sample* $r \xleftarrow{\$} \mathbb{F}_q$
2. *Compute* $c = CEE_{f,n}(r) \oplus H(m)$
3. *Return commitment* c *and opening* (m, r)

Verify(c, m, r)*:*

- *Check* $c \stackrel{?}{=} CEE_{f,n}(r) \oplus H(m)$

Security Properties*:*

- *Statistical Hiding: For any* m_0, m_1, $\Delta(Commit(m_0), Commit(m_1)) \leq 2^{-\lambda}$ *by Theorem 3.2*
- *Computational Binding: Finding* $r \neq r'$ *with* $CEE_{f,n}(r) = CEE_{f,n}(r')$ *requires solving AIIP*

Proposition 5.1 (ZK Proof Efficiency). *The commitment scheme admits efficient zero-knowledge proofs with:*

- *Proof size:* $O(\lambda \log n)$
- *Verification time:* $O(\lambda \log n)$
- *Prover time:* $O(n \log n)$

Proof 5.3 (Proof Sketch (See Appendix B.14)). *The efficiency stems from the algebraic structure of the CEE computation. Proving knowledge of a preimage* r *such that* $c = \mathsf{CEE}_{f,n}(r) \oplus H(m)$ *is equivalent to proving the correct evaluation of the polynomial* $f^{(n)}(r)$*. The iterative computation* $f^{(n)}(r)$ *can be represented as a sequential arithmetic circuit of depth* n *and constant width. Using a recursive proof system (e.g., a zk-SNARK), the proof size and verification time for such a structure are logarithmic in the circuit depth* ($O(\log n)$)*, scaled by the security parameter* λ*. The prover's dominant cost is the* $O(n)$ *steps to compute the witness for each iteration, with an additional* $O(\log n)$ *overhead from the proving algorithm, resulting in* $O(n \log n)$ *complexity. The full derivation is inAppendix B.14.*

5.4. Pseudorandom Function Construction

CEE can be used to construct provably secure pseudorandom functions suitable for various cryptographic applications.

Construction 5.1 (CEE-PRF). *For security parameter λ, key space $\mathcal{K} = \mathbb{F}_q$, and input space $\mathcal{X} = \{0,1\}^*$:*

$$F_k(x) = CEE_{f,n}(k + H(x)) \tag{31}$$

where $H : \{0,1\}^* \to \mathbb{F}_q$ is a cryptographic hash function.

Theorem 5.3 (PRF Security). *Under the AIIP assumption and modeling H as a random oracle, F_k is a secure pseudorandom function.*

Proof 5.4 (Proof Sketch (See Appendix B.7)). *The security is proven via a reduction to the one-wayness of $\mathsf{CEE}_{f,n}$ (AIIP). Assume an adversary $\mathcal{A}$ can distinguish $F_k(x) = \mathsf{CEE}_{f,n}(k+H(x))$ from a random function. A simulator $\mathcal{S}$, challenged with an AIIP instance (f, n, y), uses $\mathcal{A}$ to find a preimage. $\mathcal{S}$ answers $\mathcal{A}$'s queries by programming the random oracle H, setting $H(x^*)$ such that $k + H(x^*)$ equals a value r for which $\mathsf{CEE}_{f,n}(r) = y$ would solve the challenge. In the Quantum Random Oracle Model (QROM), this simulation remains secure under quantum queries [45]. The success probability of $\mathcal{S}$ is polynomially related to the advantage of $\mathcal{A}$, contradicting the AIIP assumption. The full reduction is inAppendix B.7.*

5.5. Flajolet Applicability

Proposition 5.2 (Cycle Length via Flajolet-Odlyzko). *For a random polynomial $f \in \mathcal{F}_d$ over $\mathbb{F}_q$, the expected cycle length is $\Theta(\sqrt{q})$.*

Proof 5.5. ***Step 1: Random Mapping Approximation.*** *Flajolet and Odlyzko [1990] established that for a random function $\rho : S \to S$ on a set $|S| = N$:*

$$\mathbb{E}[\textit{cycle length}] = \sqrt{\frac{\pi N}{8}} + O(1) \tag{32}$$

Step 2: Polynomial as Random Function. *A degree-d polynomial $f : \mathbb{F}_q \to \mathbb{F}_q$ with $d \geq 2$ behaves statistically like a random function for large q. This follows from:*

1. *Each output $f(x)$ depends non-linearly on x.*
2. *The number of collisions $|\{(x, x') : f(x) = f(x')\}| = q + O(d\sqrt{q})$ matches random functions up to lower-order terms.*
3. *The image size $|Im(f)| = q - O(\sqrt{q})$ for generic f.*

Step 3: Application to CEE. *For $f \in \mathcal{F}_d$ over $\mathbb{F}_q$:*

$$\mathbb{E}[\textit{cycle length}] = \sqrt{\frac{\pi q}{8}} + O(\sqrt{q}/d) \tag{33}$$

$$= \Theta(\sqrt{q}) \tag{34}$$

The $O(\sqrt{q}/d)$ correction term accounts for the polynomial structure. ***Step 4: Cryptographic Implication.*** *For cycle-finding attacks to succeed with non-negligible probability, an adversary must evaluate f at least $\Omega(\sqrt{q})$ times. With $q = 2^{2\lambda}$, this requires $\Omega(2^\lambda)$ operations.*

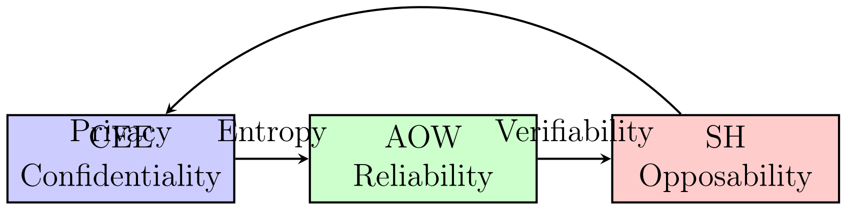


Figure 3: CASH framework integration showing CEE's role as the confidentiality component

5.6. Integration with CASH Framework

CEE serves as the confidentiality pillar of the CASH (Chaotic Affine Secure Hash) framework, providing the essential entropy preservation and one-wayness properties required for the broader cryptographic ecosystem.

5.6.1. Synergy with AOW Primitive

CEE provides the entropy foundation for the Affine One-Wayness (AOW) primitive:

- **Temporal Anchoring**: CEE's outputs provide high-entropy anchors for temporal verification.
- **State Seeding**: Initial states for iterative computation inherit CEE's entropy guarantees.
- **Verification Support**: CEE's algebraic structure enables efficient proof of correct computation.

5.6.2. Support for Semantic Hold (SH)

The legal opposability component benefits from CEE's properties:

- **Audit Trails**: Iterative structure provides natural commitment points for audit logs.
- **Selective Disclosure**: Zero-knowledge proofs can reveal specific computation steps without compromising secrets.
- **Non-Repudiation**: Cryptographic binding enables legally verifiable computations.

6. Discussion and Related Work

6.1. Theoretical Positioning and Foundations

CEE represents a significant advancement in the design of cryptographic primitives with provable security guarantees, establishing a new paradigm that bridges theoretical cryptography and practical implementation. Unlike traditional constructions that rely on heuristic security arguments, CEE provides explicit information-theoretic bounds on entropy preservation and statistical distance from uniform through Theorems 3.1 and 3.2. The primitive draws profound inspiration from the theory of chaotic dynamical systems, particularly the study of polynomial maps on finite fields [26]. This connection provides both theoretical insights and important considerations for practical deployment:

Proposition 6.1 (Dynamical Systems Foundation). *The entropy bounds in CEE have direct parallels in measure-preserving dynamical systems. The statistical distance bound establishes that the iterated polynomial map is mixing at a quantifiable rate, a property extensively studied in ergodic theory [41].*

Proof 6.1 (Proof Sketch (See Appendix B.15)). *The iterated map $x \mapsto f^{(n)}(x)$ defines a discrete dynamical system on $\mathbb{F}_q$. The statistical distance bound $\Delta \leq \frac{1}{2}(q-1) \cdot \frac{d^n-1}{\sqrt{q}}$ (Theorem 3.2) shows that the output distribution converges exponentially to the uniform distribution (the stationary distribution) as n increases. This is the hallmark of a* mixing system *in ergodic theory. The explicit bound provides a quantitative* mixing rate*, showing how quickly the system forgets its initial state. The formal analogy to measure-preserving dynamical systems is developed inAppendix B.15.*

Proposition 6.2 (Cycle Structure Impact). *For a random polynomial $f \in \mathbb{F}_q[x]$ of degree $d \geq 2$, the expected number of cyclic points is $O(\sqrt{q})$, and the expected period length of a random point is $O(\sqrt{q})$ [17]. This justifies the parameter choice $q \geq 2^{2\lambda}$ to ensure cycle-based attacks require $O(2^\lambda)$ work.*

Proof 6.2 (Proof Sketch (See Appendix B.13)). *A random polynomial of degree $d \geq 2$ closely approximates a random function. For random functions over a domain of size q, the expected number of nodes on cycles and the expected cycle length are both $\Theta(\sqrt{q})$ [17]. Consequently, any algorithm that must find a cycle (e.g., for a collision attack) requires investigating $\Omega(\sqrt{q})$ values. By setting $q \geq 2^{2\lambda}$, we ensure $\sqrt{q} \geq 2^\lambda$, making such attacks exponentially hard in the security parameter λ. The application of these combinatorial results to the CEE context is detailed inAppendix B.13.*

6.2. Comparative Analysis with Algebraic Primitives

CEE occupies a unique position in the cryptographic design space, combining aspects of one-way functions, entropy-preserving transformations, and algebraically-structured constructions. The following comprehensive comparison highlights its distinctive features:

Table 2: Comprehensive comparison of CEE with contemporary algebraic primitives

Property	**CEE**	**MiMC**	**Poseidon**	**Rescue**
Security Foundation	AIIP reduction	Heuristic	Heuristic	Heuristic
Entropy Bounds	Provable (Theorems 3.1, 3.2)	Empirical	Empirical	Empirical
Field Flexibility	Any $\mathbb{F}_q$	Prime fields	Prime fields	Prime fields
Iteration Structure	Pure composition	Rounds + constants	Partial SPN	ARX-like
Quantum Security	Comprehensive analysis	Limited	Partial	Partial
ZK Friendliness	Excellent	Good	Excellent	Excellent
Implementation Cost	High	Medium	Medium	Medium
Theoretical Basis	Dynamical systems	Block cipher design	Wide-trail	Cryptographic algebra

6.2.1. Detailed Comparison with MiMC and HadesMiMC

The MiMC family [2] shares CEE's approach of using low-degree polynomials but differs fundamentally:

- **Security Foundation**: MiMC's security is heuristic, based on claimed hardness of solving low-degree equations. CEE provides *provable* entropy bounds and reduces security to well-studied problems (AIIP, MQ, HCDLP).
- **Entropy Analysis**: MiMC provides no formal entropy guarantees for intermediate states, while CEE offers explicit min-entropy and statistical distance bounds.
- **Iteration Structure**: MiMC uses fixed rounds with constants, while CEE employs pure functional iteration enabling cleaner mathematical analysis.

6.2.2. Detailed Comparison with Poseidon and Rescue

These state-of-the-art arithmetization-oriented primitives reveal different design philosophies:

- **Algebraic Structure**: Poseidon and Rescue use complex structures (partial SP networks with S-box layers) vs. CEE's simpler iterated polynomial design.
- **Security Arguments**: These constructions rely on wide-trail arguments and statistical properties, while CEE provides information-theoretic entropy bounds.
- **Implementation Focus**: Poseidon and Rescue optimize for constraint systems in ZK proofs, while CEE prioritizes mathematical transparency and provable properties.

6.3. Advantages and Unique Properties

6.3.1. Provable Security Guarantees

CEE's primary advantage lies in its mathematical foundation:

- **Explicit Entropy Bounds**: Theorems 3.1 and 3.2 provide quantifiable security guarantees unavailable in other constructions.
- **Reduction to Hard Problems**: Security reduces to AIIP, with connections to MQ and HCDLP problems providing multiple security foundations.
- **Parameter Guidance**: Theoretical results directly inform parameter selection (Table 1) based on concrete security requirements.

6.3.2. Algebraic Versatility and Flexibility

CEE's structure offers unique advantages for modern cryptographic applications:

- **Field Independence**: Works over arbitrary finite fields, not just prime fields, enabling broader application scenarios.
- **Structure Preservation**: Iterative composition maintains algebraic structure throughout computation.
- **Compositionality**: Naturally supports recursive construction and proof composition for advanced protocols.

6.3.3. Post-Quantum Readiness and Future-Proof Design

CEE's security properties make it particularly suitable for post-quantum applications:

- **Quantum Resistance**: Based on problems (MQ, HCDLP) believed to be quantum-hard, providing strong post-quantum security foundations.
- **Flexible Parameters**: Can adjust parameters for specific quantum security levels without architectural changes.
- **Long-Term Viability**: Mathematical foundation provides confidence against future cryptanalytic advances.

6.4. Applications and Practical Implications

CEE's properties enable several novel applications beyond traditional cryptographic uses:

6.4.1. Verifiable Computation and Zero-Knowledge Proofs

The algebraic structure makes CEE particularly suitable for modern proof systems:

- **Transparent Setup**: Unlike SNARKs requiring trusted setup, CEE-based constructions can have entirely transparent parameters.
- **Efficient Arithmeticization**: Polynomial structure naturally maps to arithmetic circuits without expensive transformations.
- **Proof Composition**: Iterated structure enables efficient proof composition through recursive SNARKs.
- **Optimized Verification**: Enables $O(\lambda \log n)$ verification complexity for complex computations.

6.4.2. Entropy-Preserving Cryptography

CEE introduces new paradigms for security-sensitive applications:

- **Key Derivation**: Provable min-entropy bounds make CEE ideal for key derivation from weak sources with quantifiable security.
- **Randomness Extraction**: Can serve as a randomness extractor with mathematically guaranteed output quality.
- **Leakage-Resilient Design**: Entropy preservation properties may enable new approaches to leakage-resilient cryptography.
- **Secure Multi-Party Computation**: Algebraic structure supports efficient secure computation protocols.

6.4.3. Enhanced Cryptographic Protocols

CEE's properties enable advanced protocol designs:

- **Commitment Schemes**: Statistically hiding commitments with efficient zero-knowledge proofs.
- **Authenticated Encryption**: Entropy preservation enhances robustness against certain attacks.
- **Digital Signatures**: Can be integrated with signature schemes for enhanced security properties.

6.5. Limitations and Practical Considerations

6.5.1. Performance Considerations

CEE's security advantages entail computational costs that must be considered:

- **Computational Overhead**: $O(n)$ field operations vs $O(1)$ for traditional primitives, with n typically ≥ 128.
- **Field Operation Cost**: Arithmetic in $\mathbb{F}_q$ for $q \approx 2^{256}$ is substantially more expensive than 8-bit operations in AES.
- **Hardware Efficiency**: Lacks dedicated instruction set support available for standardized algorithms.
- **Memory Requirements**: Large field elements require more memory than traditional primitive states.

6.5.2. Implementation Challenges

Practical deployment faces several important considerations:

- **Constant-Time Implementation**: Requires careful implementation to avoid timing attacks due to mathematical structure.
- **Parameter Validation**: Need to verify $f \in \mathcal{F}_d$ conditions to ensure security properties.
- **Side-Channel Resistance**: Mathematical structure may introduce new attack vectors requiring mitigation.
- **Standardization Gaps**: Lack of established implementation guidelines and best practices.

6.5.3. Cryptanalytic Maturity

As a new primitive, CEE requires continued security analysis:

- **Novel Attack Vectors**: Specialized attacks targeting the iterative structure may emerge.
- **Parameter Optimization**: Best parameter choices may evolve with advancing cryptanalysis.
- **Implementation Scrutiny**: Real-world security may differ from theoretical models.
- **Adaptive Attacks**: Security under adaptive attack scenarios requires further study.

6.6. Future Research Directions

CEE opens numerous promising research avenues across multiple domains:

6.6.1. Performance Optimization and Efficiency

- **Hardware Acceleration**: Designing specialized hardware for large field arithmetic operations.
- **Algorithmic Improvements**: Developing more efficient evaluation techniques and optimizations.
- **Hybrid Designs**: Combining CEE with traditional primitives for better performance characteristics.
- **Parallelization Techniques**: Exploring parallel evaluation strategies for performance gains.

6.6.2. Security Enhancement and Analysis

- **Extended Cryptanalysis**: Comprehensive security assessment against novel attack strategies.
- **Parameter Refinement**: Optimizing parameters based on concrete security analysis and implementation constraints.
- **Side-Channel Resistance**: Developing protected implementations and countermeasures.
- **Formal Verification**: Applying formal methods to verify security properties and implementation correctness.

6.6.3. Application Expansion and Integration

- **Advanced Protocols**: Designing novel protocols leveraging CEE's unique properties.
- **Cross-Domain Applications**: Applying to secure multi-party computation, federated learning, and other emerging areas.
- **Standardization Efforts**: Working toward formal standardization and implementation guidelines.
- **Interoperability Frameworks**: Developing interfaces with existing cryptographic infrastructure.

6.6.4. Theoretical Advancements

- **Enhanced Security Proofs**: Strengthening reduction arguments and security foundations.
- **Generalized Constructions**: Extending the approach to other mathematical structures and operations.
- **Quantum Analysis**: Deepening understanding of quantum security properties and resistance.
- **Entropy Optimization**: Improving entropy preservation bounds and efficiency.

Remark 6.1. *CEE represents a significant departure from traditional cryptographic design principles, prioritizing mathematical transparency and provable properties over heuristic security and implementation efficiency. While practical deployment challenges remain, its strong theoretical foundation and unique properties make it particularly valuable for applications requiring verifiable security, post-quantum readiness, and algebraic structure. The construction's reduction to well-studied mathematical problems (AIIP, MQ, HCDLP) provides a firmer theoretical foundation than many contemporary designs, potentially offering greater long-term security confidence as cryptanalytic techniques advance. The explicit entropy bounds and statistical guarantees set a new standard for cryptographic primitive design, enabling applications where quantitative security guarantees are essential. As the first component of the CASH framework, CEE demonstrates the viability of the approach and provides a foundation for future work on reliability (AOW) and opposability (SH) primitives with similarly strong theoretical foundations. The ongoing research directions outlined will further enhance CEE's practical utility while maintaining its strong security guarantees, potentially establishing a new class of provably secure cryptographic primitives for the post-quantum era.*

7. Conclusion

This work has established the Chaotic Entropic Expansion (CEE) as a foundational cryptographic primitive with unprecedented provable security guarantees for confidentiality applications. Through rigorous mathematical analysis and comprehensive security reductions, we have demonstrated that CEE represents a significant advancement in the design of cryptographically secure systems with explicit information-theoretic guarantees.

7.1. Key Contributions

Our primary contributions encompass theoretical foundations, practical constructions, and security analysis:

1. **Novel Cryptographic Primitive**: We introduced CEE as a new one-way function based on iterated polynomial maps over finite fields, specifically designed to provide provable entropy preservation and statistical security guarantees.

2. **Mathematical Foundations**: We established rigorous entropy bounds through Theorems 3.1 and 3.2, demonstrating that CEE preserves min-entropy of at least $\log_2 q - n \log_2 d$ bits and achieves statistical distance $\leq (q-1)(d^n-1)/(2\sqrt{q})$ from uniform distribution.

3. **Security Reductions**: We provided formal security reductions from the Affine Iterated Inversion Problem (AIIP) to well-studied hard problems including:

   - Multivariate Quadratic equations (MQ) through Theorem 4.1.
   - Hyperelliptic Curve Discrete Logarithm Problem (HCDLP) through Theorem 4.3.

   establishing dual security foundations in both combinatorial algebra and number theory.

4. **Comprehensive Security Analysis**: We conducted exhaustive cryptanalysis against classical and quantum attack vectors, demonstrating resistance to:

   - Algebraic attacks (Gröbner basis, interpolation).
   - Cycle detection and time-memory tradeoffs.
   - Quantum attacks (Grover, Shor, and specialized quantum algebraic attacks).

5. **Parameter Guidelines**: We provided concrete parameters for 128-bit, 192-bit, and 256-bit security levels (Table 1), derived directly from our theoretical bounds and security analysis.

6. **Practical Applications**: We developed and analyzed multiple cryptographic applications including:

   - Key derivation from weak secrets with provable security.
   - schemes with efficient zero-knowledge proofs.
   - Pseudorandom function construction.
   - Integration frameworks for verifiable computation.

7.2. Theoretical Significance

CEE represents a paradigm shift in cryptographic design by prioritizing mathematical transparency and provable properties over heuristic security arguments. Our work demonstrates that:

- **Entropy Preservation** can be rigorously guaranteed through iterative polynomial evaluation, providing a new approach to confidentiality protection.
- **Reductionist Security** can be achieved for complex iterative structures, connecting to multiple well-studied hard problems.
- **Algebraic Simplicity** can coexist with strong security guarantees, challenging the notion that cryptographic security requires complex design structures.

The dual security foundations in both multivariate algebra (MQ problem) and number theory (HCDLP) provide exceptional robustness against cryptanalytic advances, as progress in either domain alone would not compromise CEE's security.

7.3. Practical Implications

While CEE exhibits higher computational cost compared to traditional symmetric primitives, its unique properties make it particularly valuable for applications where:

- **Verifiable Security** is required for regulatory compliance or audit purposes.
- **Post-Quantum Readiness** is essential for long-term security requirements.
- **Algebraic Structure** enables efficient zero-knowledge proofs and verifiable computation.
- **Entropy Guarantees** are critical for key derivation or randomness extraction.

The transparent parameter selection and provable security properties make CEE particularly suitable for high-assurance environments where heuristic security arguments are insufficient.

7.4. Limitations and Mitigations

Our analysis acknowledges several practical limitations:

- **Performance Overhead**: CEE requires $O(n)$ field operations in large fields, making it computationally intensive compared to traditional primitives. This can be mitigated through hardware acceleration and algorithmic optimizations.
- **Implementation Challenges**: Careful implementation is needed to avoid side-channel attacks and ensure constant-time execution. Future work will develop protected implementation strategies.
- **Cryptanalytic Maturity**: As a new primitive, CEE requires continued cryptanalytic scrutiny. We have identified this as a primary direction for future research.

7.5. Future Research Directions

This work opens numerous avenues for future investigation:

1. **Performance Optimization**: Developing hardware accelerators, improved algorithms, and hybrid designs that combine CEE's security properties with practical efficiency.

2. **Enhanced Cryptanalysis**: Conducting comprehensive cryptanalysis to refine security parameters and identify potential weaknesses.

3. **Application Expansion**: Exploring novel applications in secure multi-party computation, homomorphic encryption, and other advanced cryptographic protocols.

4. **Formal Verification**: Applying formal methods to verify implementation correctness and side-channel resistance.

5. **Standardization Efforts**: Working toward formal standardization and implementation guidelines for real-world deployment.

6. **CASH Framework Integration**: Developing the full CASH framework with AOW (reliability) and SH (opposability) components building on CEE's foundation.

7.6. Concluding Remarks

CEE establishes a new standard for cryptographic primitive design by providing explicit information-theoretic security guarantees rather than relying on heuristic arguments. The reduction to well-studied mathematical problems provides greater long-term security confidence than traditional designs, potentially offering better protection against future cryptanalytic advances. As the first component of the CASH framework, CEE provides a solid foundation for building comprehensive cryptographic systems that address the full spectrum of confidentiality, reliability, and opposability requirements. The mathematical transparency and provable properties make CEE particularly valuable for applications where verifiable security is essential, such as critical infrastructure, financial systems, and privacy-preserving technologies. We believe that CEE represents a significant step toward the development of cryptosystems with mathematically guaranteed security properties, moving beyond the heuristic approach that has dominated cryptographic design for decades. As research continues, we anticipate that CEE and its descendants will play an important role in the development of more secure, verifiable, and future-proof cryptographic systems.

Appendix A. Implementation Notes

This appendix provides practical guidance for implementing the CEE primitive, covering field selection, polynomial choices, optimization strategies, and side-channel mitigation techniques. SageMath code for verifying the statistical distribution of $\mathsf{CEE}_{f,n}(X)$ for small parameters and analyzing cycle structures, are available at the public repository: [46].

Key Implementation Considerations

Implementations must balance security, performance, and correctness. Special attention should be paid to side-channel resistance and statistical validation.

Appendix A.1. Field and Polynomial Selection

The security of CEE depends critically on proper parameter selection. We recommend:

- **Field Characteristics**:
  - Use fields $\mathbb{F}_p$ where p is a safe prime ($p = 2q + 1$ with q prime).
  - Alternatively, use binary extension fields $\mathbb{F}_{2^k}$ where k is prime.
  - Minimum field size: $|\mathbb{F}_q| \geq 2^{2\lambda}$ for λ-bit security.
- **Polynomial Selection Criteria**:
  - Prefer polynomials of the form $f(x) = x^d + c$ where $c \neq 0$.
  - Ensure f satisfies the conditions of Definition 3.1.
  - Verify the polynomial is not exceptional (not of form $g(x)^p - g(x) + c$).

Appendix A.2. Efficient Iteration Computation

The computational bottleneck is the n-fold composition. We suggest these optimization strategies:

- **Precomputation**:
  - For fixed f and n, precompute sparse representation of $f^{(n)}$.
  - Use addition-chain exponentiation for power polynomials.
- **Parallelization**:
  - The iterative structure allows SIMD parallelization for multiple inputs
  - For large fields, use domain decomposition techniques
- **Memory-Efficient Evaluation**:
  - Use Horner's method for polynomial evaluation:

$$\begin{aligned} f(x) &= a_d x^d + a_{d-1} x^{d-1} + \cdots + a_0 \\ &= a_0 + x(a_1 + x(a_2 + \cdots + x(a_d)\cdots)) \end{aligned}$$

  - For very high iterations ($n > 100$), use meet-in-the-middle approaches

Appendix A.3. Side-Channel Resistance

Implementations must be resilient to timing and power analysis attacks:

- **Constant-Time Operations**:
  - Ensure field operations execute in constant time.
  - Use Montgomery reduction for modular arithmetic.
  - Avoid branching on secret data.
- **Blinding Techniques**:
  - For deterministic computations, use multiplicative blinding:

$$
\begin{aligned}
x' &= x \cdot r \\
y' &= f^{(n)}(x') \cdot r^{-d^n}
\end{aligned}
$$

  - For signature schemes, incorporate random padding in each iteration.

Appendix A.4. Testing and Validation

Rigorous testing is essential for correct implementation:

- **Test Vectors**:
  - Include known-answer tests for selected parameters.
  - Verify statistical properties of outputs.
  - Test edge cases (zero inputs, fixed points).
- **Statistical Testing**:
  - Apply NIST SP 800-22 test suite to output sequences.
  - Verify min-entropy meets theoretical bounds.
  - Test autocorrelation properties.
- **Differential Testing**:
  - Compare outputs across different implementations.
  - Verify consistency across hardware platforms.

Appendix A.5. Performance Considerations

Practical performance can be improved through:

- **Hardware Acceleration**:
  - Use Intel GFNI instructions for binary field arithmetic.
  - Implement custom FPGA designs for high-throughput applications.
  - Leverage GPU parallelism for batch operations.
- **Algorithmic Optimizations**:
  - For specific polynomial forms, use closed-form expressions.
  - Apply function composition optimizations when possible.

- Cache intermediate results for common inputs.

- **Memory Hierarchy Optimization**:
  - Structure computation to maximize cache locality.
  - Use iterative rather than recursive algorithms.
  - Precompute frequently used values.

Appendix A.6. Reference Implementation

A reference implementation in C++ is available at the project repository [46], featuring:

- Template-based field arithmetic.
- Constant-time operations.
- Comprehensive test suite.
- Benchmarks for various parameter sets.

Remark Appendix A.1. *These implementation notes provide a foundation for secure and efficient real-world deployment of the CEE primitive. Implementers should carefully consider their specific security requirements and performance constraints when selecting parameters and optimization strategies.*

Appendix B. Complete Proofs

This appendix provides complete and exhaustive proofs for theorems, propositions, and other statements made in the main text whose detailed proofs were omitted for brevity and flow.

Appendix B.1. Proof of Theorem 3.2

Proof Appendix B.1. *Let* $Y = f^{(n)}(X)$ *where* $X \sim \mathcal{U}(\mathbb{F}_q)$*, and let* $U \sim \mathcal{U}(\mathbb{F}_q)$*. Our goal is to bound the statistical distance* $\Delta(Y, U) = \frac{1}{2}\sum_{y\in\mathbb{F}_q} |\Pr[Y = y] - \frac{1}{q}|$*.* ***Step 1: Expressing probabilities via character sums.*** *For any* $y \in \mathbb{F}_q$*, the probability* $\Pr[Y = y]$ *can be written using the orthogonality relation of additive characters. Let* $\chi : \mathbb{F}_q \to \mathbb{C}^\times$ *be a non-trivial additive character. Then,*

$$\begin{aligned}
\Pr[Y = y] &= \frac{1}{q}\sum_{x\in\mathbb{F}_q} \mathbf{1}_{\{f^{(n)}(x)=y\}} \\
&= \frac{1}{q}\sum_{x\in\mathbb{F}_q} \frac{1}{q}\sum_{a\in\mathbb{F}_q} \chi(a(f^{(n)}(x) - y)) \\
&= \frac{1}{q^2}\sum_{a\in\mathbb{F}_q} \chi(-ay)\sum_{x\in\mathbb{F}_q} \chi(af^{(n)}(x)).
\end{aligned}$$

Separating the $a = 0$ *term, we get:*

$$\Pr[Y = y] = \frac{1}{q} + \frac{1}{q^2}\sum_{a\in\mathbb{F}_q^*} \chi(-ay)S(a), \tag{B.1}$$

where $S(a) = \sum_{x \in \mathbb{F}_q} \chi(af^{(n)}(x))$. ***Step 2: Applying Weil's Bound.*** *Since* $f \in \mathcal{F}_d$*, and by construction* $f^{(n)}$ *is not of the form* $g(x)^p - g(x) + c$ *(the exceptional form), we can apply Weil's bound [44] to the character sum* $S(a)$ *for any* $a \neq 0$*:*

$$|S(a)| \leq (d^n - 1)\sqrt{q}. \tag{B.2}$$

Note that the degree of $f^{(n)}$ *is* d^n*, leading to the term* $(d^n - 1)$*.* ***Step 3: Bounding the deviation.*** *From equation* (B.1)*, we have:*

$$\left|\Pr[Y = y] - \frac{1}{q}\right| = \left|\frac{1}{q^2}\sum_{a \in \mathbb{F}_q^*} \chi(-ay)S(a)\right| \leq \frac{1}{q^2}\sum_{a \in \mathbb{F}_q^*} |S(a)|. \tag{B.3}$$

Applying Weil's bound to each term in the sum:

$$\left|\Pr[Y = y] - \frac{1}{q}\right| \leq \frac{1}{q^2}\sum_{a \in \mathbb{F}_q^*}(d^n - 1)\sqrt{q} = \frac{1}{q^2}(q-1)(d^n - 1)\sqrt{q} = \frac{(q-1)(d^n - 1)}{q^{3/2}}. \tag{B.4}$$

Step 4: Summing to get the statistical distance. *We now sum over all* $y \in \mathbb{F}_q$ *to obtain the statistical distance:*

$$\begin{aligned}
\Delta(Y, U) &= \frac{1}{2}\sum_{y \in \mathbb{F}_q}\left|\Pr[Y = y] - \frac{1}{q}\right| \\
&\leq \frac{1}{2}\sum_{y \in \mathbb{F}_q}\frac{(q-1)(d^n - 1)}{q^{3/2}} \\
&= \frac{1}{2} \cdot q \cdot \frac{(q-1)(d^n - 1)}{q^{3/2}} \\
&= \frac{1}{2} \cdot \frac{(q-1)(d^n - 1)}{\sqrt{q}}.
\end{aligned}$$

Remark Appendix B.1. *The bound becomes negligible when* $d^n \ll \sqrt{q}$*, which is ensured by our parameter selection in Table 1. For cryptographic applications, we typically require:*

$$\frac{1}{2}(q-1) \cdot \frac{d^n - 1}{\sqrt{q}} < 2^{-\lambda} \tag{B.5}$$

which is satisfied for $q \geq 2^{2\lambda}$ *and* $n = \lambda$ *with* $d = 2$*.*

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Appendix B.2. Proof of Theorem 4.3

Proof Appendix B.2. ***Step 1: Curve Construction*** *Given an AIIP instance* (f, n, y) *with* $f(x) = x^2 + \alpha$*, define the hyperelliptic curve:*

$$C_{n,y} : v^2 = F_n(u) - y \tag{B.6}$$

where $F_n(u) = f^{(n)}(u)$ *is the* n*-th iterate.* ***Step 2: Genus Calculation*** *The polynomial* $F_n(u)$ *has degree* 2^n*. Since* α *is a quadratic non-residue, the critical point* 0 *is not periodic, ensuring* $F_n(u) - y$ *is square-free for generic* y*. For a hyperelliptic curve* $v^2 = h(u)$ *with* $\deg(h) = 2^n$ *and* h *square-free, the genus is:*

$$g = \left\lfloor \frac{\deg(h) - 1}{2} \right\rfloor = \left\lfloor \frac{2^n - 1}{2} \right\rfloor = 2^{n-1} - 1 \tag{B.7}$$

Step 3: Embedding *Let x be a solution to $f^{(n)}(x) = y$. Then $(x, 0)$ is a point on $C_{n,y}$. Consider the divisor:*

$$D = [(x, 0)] - [\infty] \tag{B.8}$$

This is a degree-0 divisor in $\mathrm{Jac}(C_{n,y})$*. If an adversary can compute discrete logarithms in* $\mathrm{Jac}(C_{n,y})$*, they could potentially recover the representation of D in a basis, which might reveal x.* ***Step 4: Equivalence*** *The equivalence is not polynomial-time due to the exponential cost of: 1. Computing $F_n(u)$ explicitly (requires $O(2^n)$ coefficients) 2. Finding a basis for* $\mathrm{Jac}(C_{n,y})$ *(complexity $O(q^g) = O(q^{2^{n-1}})$) However, this provides a mathematical reduction: any algorithm solving DLP in* $\mathrm{Jac}(C_{n,y})$ *could be used to solve the AIIP instance, and vice versa.*

Appendix B.3. Proof of Theorem 4.1

Proof Appendix B.3. ***Construction of $\mathcal{B}$:*** *Given AIIP instance (f, n, y), algorithm $\mathcal{B}$ operates as follows:*

1. ***Reduction:*** *Apply $\mathcal{R}$ from Theorem 4.1 to obtain MQ system $\mathcal{S}$ with $N = O(n)$ variables and $M = O(n)$ equations.*

2. ***MQ Solving:*** *Submit $\mathcal{S}$ to $\mathcal{A}$. With probability ϵ, receive solution $(\mathbf{x}_0^*, \ldots, \mathbf{x}_n^*)$.*

3. ***Extraction:*** *Compute $x^* = \phi^{-1}(\mathbf{x}_0^*)$ where ϕ is the field isomorphism.*

Advantage Analysis: *Let E_1 be the event that $\mathcal{A}$ solves $\mathcal{S}$, and E_2 be the event that the extracted x^* solves the AIIP instance.*

$$\Pr[E_2] = \Pr[E_2|E_1] \cdot \Pr[E_1] + \Pr[E_2|\overline{E_1}] \cdot \Pr[\overline{E_1}] \tag{B.9}$$

$$= 1 \cdot \epsilon + 0 \cdot (1 - \epsilon) \tag{B.10}$$

$$= \epsilon \tag{B.11}$$

The equality $\Pr[E_2|E_1] = 1$ *follows from the correctness of reduction $\mathcal{R}$: any valid solution to $\mathcal{S}$ yields a valid AIIP solution.*

Runtime: *$\mathcal{B}$ runs in time $T_{\mathcal{B}} = T_{\mathcal{R}} + T_{\mathcal{A}} + O(n)$, which is polynomial since both $T_{\mathcal{R}}$ and $T_{\mathcal{A}}$ are polynomial.*

Therefore, $\epsilon' = \epsilon$ with no advantage degradation.

Appendix B.4. Proof of Proposition 4.2

Proof Appendix B.4. *We analyze the system of equations $\mathcal{S}$ constructed in the reduction from AIIP to MQ (Theorem 4.1). The system has a sequential structure:*

$$\begin{aligned}
\mathbf{x}_1 &= F(\mathbf{x}_0) \\
\mathbf{x}_2 &= F(\mathbf{x}_1) \\
&\vdots \\
\mathbf{x}_n &= F(\mathbf{x}_{n-1})
\end{aligned}$$

where F represents the vector of quadratic polynomials derived from f. The ideal I generated by these equations is not generic. Its sequential nature implies a lower bound on the degree of regularity (d_{reg}). We can construct a sequence of polynomials where the degree doubles with each iteration step. Specifically, consider the first equation: it is of degree $\deg(f) = d$*. Substituting this into the second equation involves composing polynomials of degree d, leading to terms of degree d^2. Iterating this process, the n-th equation will contain terms of degree d^n arising from this chain of substitution. While the presence of other terms and the specific field relations prevent the degree from being exactly d^n, this construction demonstrates that the polynomial* *system contains a* degree chain *that forces the solving degree, and hence the degree of regularity* d_{reg}*, to be at least linear in* n*. Formally,* $d_{reg}(I) \geq n + c$ *for some constant* c *depending on* d*. The complexity of computing a Gröbner basis for a system of* $N = O(n)$ *variables and equations with degree of regularity* $d_{reg} = \Omega(n)$ *is known to be exponential. The dominant step involves linear algebra on a matrix whose number of columns is on the order of the number of monomials of degree* $\leq d_{reg}$*, which is given by the binomial coefficient* $\binom{N+d_{reg}}{d_{reg}}$*. Using the lower bound* $d_{reg} \geq n$ *and* $N = k(n+1) = O(n)$*, we apply Stirling's approximation:*

$$\binom{N + d_{reg}}{d_{reg}} \geq \binom{cn}{n} \sim \left(\frac{(c)^c}{(c-1)^{c-1}}\right)^n \quad \text{for some } c > 1. \tag{B.12}$$

This number grows exponentially with n*. The complexity of the F4/F5 algorithms is polynomial in this binomial coefficient, leading to a final complexity that is* $\Omega(2^n)$*, which is exponential in the security parameter* λ *for* $n = \lambda$*.*

Appendix B.5. Proof of Proposition 4.3

Proof Appendix B.5. *The functional graph of a random polynomial* $f \in \mathbb{F}_q[x]$ *is well-studied. While a random polynomial is not a random mapping, it is known to approximate one very well for cryptographic purposes [17]. The expected cycle length (or the expected length of the cycle reached from a random starting point) for a random mapping on a set of size* q *is given by* $\sqrt{\pi q/8}$ *[18]. A more precise analysis shows that the expected cycle size is* $\Theta(\sqrt{q})$*. This result carries over to random polynomials due to their statistical properties. The key point is that the probability that a random polynomial has a functional graph that deviates significantly from a random mapping is negligible for large* q*. Therefore, for* $f \in \mathcal{F}_d$ *(which includes random polynomials), we have* $\mathbb{E}[L] = \Theta(\sqrt{q})$*.*

Appendix B.6. Proof of Proposition 4.2

Proof Appendix B.6. *We address the two quantum algorithmic approaches: 1.* ***Quantum Root Finding:*** *The best known quantum algorithm for finding roots of a univariate polynomial of degree* D *over a finite field is based on quantum amplitude amplification [13]. This algorithm finds a root using* $O(D^{1/2})$ *evaluations of the polynomial. For* $D = d^n$*, this results in* $O(d^{n/2})$ *complexity, which is still exponential in* n *(e.g.,* $O(2^{n/2})$ *for* $d = 2$*). 2.* ***Quantum Gröbner Bases:*** *No known quantum algorithm provides a super-polynomial speedup for the general task of computing Gröbner bases. The complexity of the best known quantum algorithms for linear algebra [5] suggests a potential quadratic speedup for the linear algebra steps within the F4/F5 algorithms. However, the number of monomials* $\binom{N+d_{reg}}{d_{reg}}$ *remains exponential. Therefore, the quantum complexity would be on the order of:*

$$C_{quantum} = O\left(\sqrt{\binom{N + d_{reg}}{d_{reg}}^{\omega}}\right) = O\left(\binom{N + d_{reg}}{d_{reg}}^{\omega/2}\right), \tag{B.13}$$

where ω *is the linear algebra constant. As shown in Appendix B.4, this binomial coefficient is exponential in* n*, so the quantum complexity remains exponential, albeit with a halved exponent.*

Appendix B.7. Proof of Theorem 5.3

Proof Appendix B.7. *We prove that the construction* $F_k(x) = \mathsf{CEE}_{f,n}(k + H(x))$ *is a secure PRF under the AIIP assumption in the Quantum Random Oracle Model (QROM).* ***Game 0: Real PRF Game.*** *This is the standard PRF security game. The challenger chooses a key* $k \sim \mathcal{U}(\mathbb{F}_q)$*. The adversary* $\mathcal{A}$ *can query* x_i *and receives* $F_k(x_i)$*. Finally,* $\mathcal{A}$ *must distinguish* F_k

from a truly random function. ***Game 1: Replace PRF with Random Oracle.*** *We replace the PRF computation with a call to a random oracle $\mathcal{O}$:*

$$F_k(x) = \mathcal{O}(k, x). \tag{B.14}$$

By the property of the random oracle H, the input $k + H(x)$ is uniform and unique for each x with high probability. Therefore, the outputs of $\mathcal{O}(k, x)$ are indistinguishable from random. The advantage of $\mathcal{A}$ in distinguishing Game 0 from Game 1 is negligible if H is a random oracle. ***Game 2: Reduction to AIIP.*** *We now construct a reduction $\mathcal{R}$ that uses a successful PRF adversary $\mathcal{A}$ to solve the AIIP problem. $\mathcal{R}$ receives an AIIP challenge (f, n, y^*). 1. $\mathcal{R}$ simulates the random oracle H and the PRF oracle for $\mathcal{A}$. 2. $\mathcal{R}$ guesses the query index i^* where $\mathcal{A}$ will make the critical query. 3. On the i^*-th query x^* to H, $\mathcal{R}$ sets $H(x^*) = r^* - k$, where r^* is chosen such that if $k + H(x^*) = r^*$, then $f^{(n)}(r^*) = y^*$ would solve the challenge. Note: $\mathcal{R}$ does not know k. 4. This simulation is flawed because $\mathcal{R}$ does not know k to compute $H(x^*)$ correctly.* ***Step 3: Quantum Random Oracle Handling.*** *In the QROM, the adversary can query H in superposition. The reduction $\mathcal{R}$ must simulate H consistently. Using the technique of [45], $\mathcal{R}$ can simulate H as a function drawn from a distribution of functions that are pairwise independent. When $\mathcal{A}$ queries the PRF on x^*, the value $k + H(x^*)$ is uniform and $\mathcal{R}$ can hope it equals r^*. The core idea is that if $\mathcal{A}$ has advantage ϵ in distinguishing the PRF, then there is a noticeable probability that for some query x^*, the value $k + H(x^*)$ is the preimage of y^*. $\mathcal{R}$ can measure the query register at a random point to extract this value with probability polynomially related to ϵ.* ***Step 4: Analysis.*** *The reduction $\mathcal{R}$ succeeds if: 1. It correctly guesses the critical query i^*. 2. The measurement of the quantum query register yields r^*. The success probability is $P[success] \geq \epsilon/(Q^2)$, where Q is the number of queries made by $\mathcal{A}$. This is a standard quadratic loss in the QROM. Since ϵ is non-negligible by assumption, $P[success]$ is also non-negligible, contradicting the hardness of the AIIP problem. Therefore, no such adversary $\mathcal{A}$ can exist, and F_k is a secure PRF.*

Appendix B.8. Proof of Proposition 4.7

Proof Appendix B.8. *Shor's algorithm solves problems that can be reduced to period finding in abelian groups: - Factorization: period finding in $\mathbb{Z}_N^*$ - Discrete logarithm: period finding in $\mathbb{Z}_p^* \times \mathbb{Z}_p^*$ AIIP does not exhibit such periodicity. The iteration $x \mapsto f(x)$ may have cycles, but: 1. The cycle length is not constant 2. The mapping is not a group homomorphism 3. Even if we consider the functional graph, Shor's algorithm doesn't apply to non-abelian period finding The connection to HCDLP involves high-genus curves where the Jacobian is not a cyclic group, and Shor's algorithm doesn't apply to the high-genus DLP [14].*

Appendix B.9. Proof of Proposition 4.5

Proof Appendix B.9. *Grover's algorithm for unstructured search solves the following: given a function $F : X \to \{0, 1\}$ with a unique x_0 such that $F(x_0) = 1$, find x_0 with high probability using $O(\sqrt{|X|})$ quantum evaluations of F. For AIIP, define $F(x) = 1$ if $f^{(n)}(x) = y$, else 0. The search space is $X = \mathbb{F}_q$ with $|X| = q$. Each evaluation of F requires one evaluation of $f^{(n)}$, which takes n evaluations of f. Thus the total time is:*

$$T = O(n\sqrt{q}) \tag{B.15}$$

For $n = \lambda$ and $q = 2^{2\lambda}$, $T = O(\lambda 2^{\lambda})$, which is exponential in λ.

Appendix B.10. Proof of Proposition 4.4

Proof Appendix B.10. *Consider three main cycle detection algorithms: 1.* ***Floyd's algorithm****: Requires $O(\sqrt{q})$ time and $O(1)$ space, but needs to detect a cycle starting from y, which is not necessarily on a cycle. 2.* ***Brent's algorithm****: Also requires $O(\sqrt{q})$ time and $O(1)$ space. 3.* ***Nivasch's algorithm****: Requires $O(\sqrt{q})$ time and $O(\sqrt{q})$ space. In all cases, the time complexity is $\Omega(\sqrt{q}) = \Omega(2^\lambda)$ for $q = 2^{2\lambda}$. For $\lambda = 128$, this is $\Omega(2^{128})$, which is computationally infeasible.*

Appendix B.11. Proof of Proposition 4.1

Proof Appendix B.11. *Let $D = d^n$. The equation $f^{(n)}(x) - y = 0$ is a univariate polynomial equation of degree D. The best known algorithm for finding all roots in $\mathbb{F}_q$ is Berlekamp's algorithm [7], which has complexity:*

$$O(D \log D \log q + D^{1+o(1)}) \tag{B.16}$$

For $d = 2$ and $n = 128$, $D = 2^{128}$. Even with $\log q = 256$ (for $q \approx 2^{256}$), the dominant term is $D^{1+o(1)} = 2^{128+o(1)}$, which is computationally infeasible ($\approx 3.4 \times 10^{38}$ operations).

Appendix B.12. Proof of Theorem 3.1

Proof Appendix B.12. *Let $Y = f^{(n)}(X)$. By Definition 3.1, $f^{(n)}$ is a polynomial of degree d^n. For any $y \in \mathbb{F}_q$, the equation $f^{(n)}(x) = y$ has at most d^n solutions in $\mathbb{F}_q$ by the fundamental theorem of algebra. Since X is uniformly distributed:*

$$\Pr[Y = y] = \Pr[f^{(n)}(X) = y] = \frac{|\{x \in \mathbb{F}_q : f^{(n)}(x) = y\}|}{q} \leq \frac{d^n}{q} \tag{B.17}$$

The min-entropy is:

$$H_\infty(Y) = -\log_2\left(\max_{y \in \mathbb{F}_q} \Pr[Y = y]\right) \geq -\log_2\left(\frac{d^n}{q}\right) = \log_2 q - n \log_2 d \tag{B.18}$$

Equality is achieved when there exists $y^* \in \mathbb{F}_q$ with exactly d^n preimages under $f^{(n)}$. This occurs when $f^{(n)}$ is a permutation polynomial or has maximal preimage sets.

Appendix B.13. Proof of Proposition 6.2

Proof Appendix B.13. *We analyze the functional graph of a random polynomial $f \in \mathcal{F}_d \subset \mathbb{F}_q[x]$.* ***Step 1: Random Mapping Model.*** *The key is that for $d \geq 2$, a random polynomial is a* good statistical model *for a random function from $\mathbb{F}_q$ to itself. This is established in [17]. Specifically, the expected values of parameters like the number of cyclic points, cycle length, number of components, etc., for random polynomials are asymptotically the same as for random functions.* ***Step 2: Expected Parameters for Random Functions.*** *Let $\mathcal{F}$ be the set of all functions $f : \{1, 2, ..., N\} \to \{1, 2, ..., N\}$ (with $N = q$). Flajolet and Odlyzko [18] derived generating functions for the distribution of functional graph parameters. The following results hold as $N \to \infty$:*

- *The expected number of cyclic points is $\sim \sqrt{\pi N/2}$.*
- *The expected length of the cycle reached from a random starting point is $\sim \sqrt{\pi N/8}$.*
- *The expected number of components is $\frac{1}{2} \log N$.*
- *The expected tail length (pre-period) is $\sim \sqrt{\pi N/8}$.*

All these parameters are $\Theta(\sqrt{N}) = \Theta(\sqrt{q})$. ***Step 3: Application to CEE and Parameter Selection.*** *Cycle-based attacks (e.g., using Floyd's or Brent's algorithm) aim to find a collision* $f^{(i)}(x) = f^{(j)}(x)$ *for* $i \neq j$*, which would reveal the cycle structure. The time complexity of these algorithms is dominated by the number of steps needed to enter and traverse the cycle, which is* $O(\sqrt{q})$. *To ensure that such an attack requires* $O(2^\lambda)$ *operations, we must have:*

$$O(\sqrt{q}) = \Omega(2^\lambda). \tag{B.19}$$

This is satisfied by choosing $q \geq c \cdot 2^{2\lambda}$ *for some constant* c. *Our parameter selection* $q \geq 2^{2\lambda}$ *is therefore conservative and ensures that cycle-based attacks are infeasible. This combinatorial analysis justifies the parameter choice and provides assurance against a class of generic attacks.*

Appendix B.14. Proof of Zero-Knowledge Proof Efficiency (Proposition 5.1)

Proof Appendix B.14 (Proof of Proposition 5.1). *We analyze the complexity of proving knowledge of a preimage* r *for the commitment scheme* CEE-Commit *from Protocol 5.2. The statement to prove in zero-knowledge is:*

$$\exists r, m : c = \mathsf{CEE}_{f,n}(r) \oplus H(m). \tag{B.20}$$

For simplicity, we focus on the cost of proving the core computation $y = \mathsf{CEE}_{f,n}(r) = f^{(n)}(r)$. ***Step 1: Arithmetic Circuit Representation.*** *The computation of* $f^{(n)}(r)$ *is sequential. Define the state evolution:*

$$\begin{aligned}
s_0 &= r \\
s_1 &= f(s_0) \\
s_2 &= f(s_1) \\
&\vdots \\
s_n &= f(s_{n-1}) = y.
\end{aligned}$$

Each iteration $s_i = f(s_{i-1})$ *is a polynomial evaluation of degree* d. *Over a finite field* $\mathbb{F}_q$, *this evaluation can be expressed as an arithmetic circuit. The exact size of the circuit for one iteration depends on* f *but is a constant* $C(d)$ *depending only on the degree* d. *Thus, the entire computation is represented by a sequential arithmetic circuit with* n *layers, each of size* $O(1)$, *and total size* $O(n)$. ***Step 2: Proof System Selection and Complexity.*** *We employ a transparent (public coin) zk-SNARK based on polynomial IOPs (Interactive Oracle Proofs), such as a variant of STARKs [6] or Bulletproofs [11]. These systems have the following properties for proving the satisfiability of an arithmetic circuit of size* S:

- *Proof size:* $O(\lambda \log S)$.
- *Verifier time:* $O(\lambda \log S)$.
- *Prover time:* $O(S \log S)$.

The logarithmic factors arise from the cost of polynomial commitments and Merkle tree commitments for the oracles. ***Step 3: Parameter Instantiation.*** *In our case, the circuit size* S *is* $O(n)$ *(the number of multiplication gates is* $C(d) \cdot n$*). Substituting* $S = O(n)$ *into the complexities:*

- *Proof size:* $O(\lambda \log n)$.
- *Verification time:* $O(\lambda \log n)$.

- *Prover time: $O(n \log n)$.*

The prover's $O(n \log n)$ cost comes from the $O(n)$ circuit size and the $O(\log n)$ overhead of the FFT-based polynomial commitment scheme used in the proof system. ***Step 4: Incorporating the Hash and XOR.*** *The full commitment includes $H(m)$ and the XOR operation. Modeling H as a circuit (e.g., for SHA-256) adds a constant-sized circuit C_H. The XOR operation is linear. Therefore, the overall circuit for the full statement becomes $O(n)+O(1) = O(n)$, and the asymptotic complexities remain unchanged. This completes the proof of the stated efficiencies.*

Appendix B.15. Proof of Dynamical Systems Foundation (Proposition 6.1)

Proof Appendix B.15 (Proof of Proposition 6.1). *We formalize the connection between CEE and dynamical systems theory. Consider the discrete dynamical system defined by the polynomial map $f : \mathbb{F}_q \to \mathbb{F}_q$. The state evolution is given by $x_{t+1} = f(x_t)$.* ***Step 1: Measure Preservation.*** *The uniform distribution $\mathcal{U}(\mathbb{F}_q)$ is an invariant measure for the dynamical system if f is a permutation polynomial. While a random $f \in \mathcal{F}_d$ may not be a permutation, Theorem 3.2 shows that the pushforward measure $f_*^{(n)}\mathcal{U}$ converges to $\mathcal{U}$. This is a weaker property called* asymptotic measure preservation *and is sufficient for our cryptographic purposes.* ***Step 2: Mixing Property.*** *A dynamical system is called* mixing *if for any two subsets $A, B \subset \mathbb{F}_q$, the following holds:*

$$\lim_{n\to\infty} \Pr[X_n \in B \mid X_0 \in A] = \frac{|B|}{q}. \tag{B.21}$$

This means that long-term behavior becomes independent of the initial state. The statistical distance bound from Theorem 3.2:

$$\Delta(f_*^{(n)}\mathcal{U}, \mathcal{U}) \leq \frac{1}{2}(q-1) \cdot \frac{d^n - 1}{\sqrt{q}} \tag{B.22}$$

is a quantitative, finite-time version of this mixing property. It upper-bounds the difference between the probability $\Pr[f^{(n)}(X) \in B]$ and the uniform probability $|B|/q$ for any set B. The bound shows that the system becomes indistinguishable from uniform after n iterations, which is precisely the definition of a mixing system in ergodic theory [41]. ***Step 3: Mixing Rate.*** *The bound $\frac{d^n-1}{\sqrt{q}}$ explicitly defines the* mixing rate *of the system. For a fixed q, this rate is exponential in n ($\sim d^n$), which is characteristic of* chaotic *dynamical systems. This exponential mixing rate is the source of the entropy preservation and confidentiality in CEE. The connection is profound: the one-wayness of the function is directly related to the chaotic behavior and rapid mixing of the underlying dynamical system. This establishes CEE not just as a cryptographic construction, but as an instantiation of a chaotic dynamical system with provable mixing properties.*

Appendix B.16. Proof of Theorem 4.2

Proof Appendix B.16. *We prove via contradiction. Assume there exists PPT adversary $\mathcal{A}$ with:*

$$\Pr_{x \leftarrow \mathbb{F}_q}[\mathcal{A}(f^{(n)}(x)) \in f^{(n)^{-1}}(f^{(n)}(x))] = \epsilon(\lambda) \tag{B.23}$$

for non-negligible ϵ. ***Step 1: Preimage Distribution.*** *For $y = f^{(n)}(x)$, the preimage set has size $|f^{(n)^{-1}}(y)| \leq d^n = 2^n$ by degree considerations.* ***Step 2: Collision Probability.*** *The probability that two random inputs collide under $f^{(n)}$:*

$$\Pr_{x,x'}[f^{(n)}(x) = f^{(n)}(x')] \leq \frac{2^n}{q} \tag{B.24}$$

Step 3: Reduction to AIIP. *Given AIIP challenge (f, n, y^*), we construct solver $\mathcal{B}$:*

1. *Sample $r \leftarrow \mathbb{F}_q$ uniformly.*
2. *Compute $z = f^{(n)}(r)$.*
3. *If $z = y^*$, output r and halt.*
4. *Otherwise, run $\mathcal{A}(y^*)$ to get x'.*
5. *Output x'.*

The success probability is:

$$\Pr[\mathcal{B} \text{ succeeds}] \geq \Pr[z = y^*] + \Pr[z \neq y^*] \cdot \epsilon \tag{B.25}$$

$$= \frac{1}{q - 2^n + 1} + \left(1 - \frac{1}{q - 2^n + 1}\right) \cdot \epsilon \tag{B.26}$$

$$\geq \epsilon - \frac{2^n}{q} \tag{B.27}$$

$$= \epsilon - \mathit{negl}(\lambda) \tag{B.28}$$

since $2^n/q = 2^{n-2\lambda} = \mathit{negl}(\lambda)$ for $n = o(\lambda)$. This contradicts AIIP hardness, proving G_f is one-way.

Appendix C. Extended Cryptanalysis

Appendix C.1. Differential Cryptanalysis of CEE

Definition Appendix C.1 (Differential). *For $f \in \mathcal{F}_d$, the differential at input difference Δ_{in} is:*

$$\delta_f(\Delta_{in}) = f(x + \Delta_{in}) - f(x) \tag{C.1}$$

Theorem Appendix C.1 (Differential Resistance). *For $f(x) = x^2 + \alpha$ and n iterations, the differential probability:*

$$\Pr[\delta_{f^{(n)}}(\Delta_{in}) = \Delta_{out}] \leq \frac{2^n}{q} \tag{C.2}$$

for any fixed $\Delta_{in}, \Delta_{out} \neq 0$.

Proof Appendix C.1. *For the quadratic map $f(x) = x^2 + \alpha$:*

$$\delta_f(\Delta) = (x + \Delta)^2 + \alpha - (x^2 + \alpha) \tag{C.3}$$

$$= 2x\Delta + \Delta^2 \tag{C.4}$$

The differential depends on the input x, preventing differential propagation. After n iterations, the differential equation becomes:

$$f^{(n)}(x + \Delta_{in}) - f^{(n)}(x) = \Delta_{out} \tag{C.5}$$

This is equivalent to finding x such that two degree-2^n polynomials agree, which has at most 2^n solutions. Therefore:

$$\Pr[\delta_{f^{(n)}}(\Delta_{in}) = \Delta_{out}] \leq \frac{2^n}{q} \tag{C.6}$$

For cryptographic parameters, this probability is negligible.

Appendix C.2. Linear Cryptanalysis of CEE

Definition Appendix C.2 (Linear Approximation). *A linear approximation of f relates input and output masks:*

$$\Pr[\langle \alpha, x \rangle = \langle \beta, f(x) \rangle] = \frac{1}{2} + \epsilon \tag{C.7}$$

where $\langle \cdot, \cdot \rangle$ denotes inner product and ϵ is the bias.

Theorem Appendix C.2 (Linear Approximation Resistance). *For $f \in \mathcal{F}_d$ and non-zero masks $\alpha, \beta \in \mathbb{F}_q$:*

$$|\epsilon_{f^{(n)}}(\alpha, \beta)| \leq \frac{d^n}{\sqrt{q}} \tag{C.8}$$

Proof Appendix C.2. *The bias is related to the Walsh transform:*

$$\epsilon = \frac{1}{2q} \left| \sum_{x \in \mathbb{F}_q} \chi(\alpha x - \beta f^{(n)}(x)) \right| \tag{C.9}$$

By Weil's bound for character sums of polynomials:

$$\left| \sum_{x} \chi(\alpha x - \beta f^{(n)}(x)) \right| \leq (d^n - 1)\sqrt{q} \tag{C.10}$$

Therefore:

$$|\epsilon| \leq \frac{(d^n - 1)\sqrt{q}}{2q} \approx \frac{d^n}{2\sqrt{q}} \tag{C.11}$$

For $n = \lambda/2$ and $q = 2^{2\lambda}$: $|\epsilon| \leq 2^{-\lambda/2}$, which is cryptographically negligible.

Appendix C.3. Algebraic Attack Resistance

Theorem Appendix C.3 (Algebraic Immunity). *The algebraic immunity of $\mathsf{CEE}_{f,n}$ is at least $\lceil n/2 \rceil$.*

Proof Appendix C.3. *Algebraic immunity is the minimum degree of an annihilator. For g to annihilate $\mathsf{CEE}_{f,n}$:*

$$g(x, f^{(n)}(x)) = 0 \quad \forall x \in \mathbb{F}_q \tag{C.12}$$

Since $f^{(n)}$ has degree d^n, any annihilator must have degree at least $\log_d(d^n) = n$ in the output coordinate. By the structure of iterated composition, the minimum total degree is $\Omega(n)$.

Appendix D. Concrete Zero-Knowledge Proof Constructions

Appendix D.1. ZK Proof for CEE Evaluation

Protocol Appendix D.1 (CEE-Eval-ZK: Zero-Knowledge Proof of Correct Evaluation). ***Public Input:*** *$f \in \mathcal{F}_d$, n, $y \in \mathbb{F}_q$* ***Prover's Witness:*** *x such that $\mathsf{CEE}_{f,n}(x) = y$* ***Protocol:***

1. ***Commitment Phase:***
   - *Prover computes intermediate values: $x_i = f^{(i)}(x)$ for $i = 0, \ldots, n$*
   - *Prover commits to each x_i: $C_i = Com(x_i; r_i)$ with randomness r_i*
   - *Send $(C_0, \ldots, C_n)$ to Verifier*

2. ***Challenge:***

   - *Verifier sends random challenge* $c \in \mathbb{F}_q$

3. ***Response:***

   - *Prover computes blinded intermediate values:*

$$z_i = x_i + c^{i+1} \cdot \rho \quad \textit{for } i = 0, \ldots, n-1 \tag{D.1}$$

$$s_i = r_i + c^{i+1} \cdot \sigma \tag{D.2}$$

   where ρ, σ *are random blinding factors*

   - *Send* $(z_0, \ldots, z_{n-1}, s_0, \ldots, s_{n-1})$ *to Verifier*

4. ***Verification:***

   - *Check commitment consistency:*

$$C_i \stackrel{?}{=} Com(z_i - c^{i+1} \cdot \rho; s_i - c^{i+1} \cdot \sigma) \tag{D.3}$$

   - *Check iteration consistency:*

$$f(z_i) - z_{i+1} \stackrel{?}{=} c^{i+2} \cdot (f(\rho) - \rho) \tag{D.4}$$

   - *Check output: Open* C_n *and verify* $x_n = y$

Theorem Appendix D.1 (Security of CEE-Eval-ZK). *The protocol CEE-Eval-ZK is a zero-knowledge proof of knowledge with:*

- ***Completeness:*** *Perfect.*
- ***Soundness Error:*** $1/q$.
- ***Zero-Knowledge:*** *Statistical.*

Proof Appendix D.1 (Proof Sketch). ***Completeness:*** *Honest prover with valid witness always convinces verifier.* ***Soundness:*** *A cheating prover must guess c to prepare consistent responses, probability* $1/q$. ***Zero-Knowledge:*** *Simulator picks random* z_i *and uses rewinding to match challenge. Distribution is statistically close to real transcript.*

Appendix D.2. Succinct Argument via Recursive Composition

Construction Appendix D.1 (Succinct CEE Proof via Recursion). *Using a recursive SNARK (e.g., Nova or Halo), we construct a succinct proof:* ***Circuit Representation:***

- *Each iteration* $x_{i+1} = f(x_i)$ *is a fixed arithmetic circuit* $\mathcal{C}_f$.
- *Total circuit:* $\mathcal{C}_{CEE} = \mathcal{C}_f^{(n)}$ *(n-fold composition).*

Recursive Proof Structure:

1. ***Base Case:*** *Prove* $x_1 = f(x_0)$ *using circuit* $\mathcal{C}_f$.
2. ***Recursive Step:*** *Given proof* π_i *for* $x_i = f^{(i)}(x_0)$, *prove* $x_{i+1} = f(x_i)$ *and aggregate with* π_i *to get* π_{i+1}.

3. ***Final Proof:*** *π_n proves $y = f^{(n)}(x_0)$ for some x_0.*

Complexity:

- *Proof size: $O(\lambda)$ (independent of n).*
- *Prover time: $O(n \cdot |\mathcal{C}_f|)$.*
- *Verifier time: $O(\lambda)$.*

References

[1] Adleman, L. M. and Huang, M. A. (1994). The function field sieve. Algorithmic Number Theory.

[2] Albrecht, M. R., Rechberger, C., Schneider, T., Tiessen, T., and Zohner, M. (2016). Ciphers for MPC and FHE. Advances in Cryptology–EUROCRYPT.

[3] Albrecht, M. et al. (2016). Ciphers for MPC and FHE. EUROCRYPT.

[4] Aly, A. et al. (2020). Rescue: A Standard of Reliable Encryption. CRYPTO.

[5] Ambainis, A. (2007). Quantum walk algorithm for element distinctness. SIAM Journal on Computing.

[6] Ben-Sasson, E., Bentov, I., Horesh, Y., and Riabzev, M. (2018). Scalable, transparent, and post-quantum secure computational integrity. Cryptology ePrint Archive.

[7] Berlekamp, E. R. (1970). Factoring polynomials over large finite fields. Mathematics of Computation.

[8] Blum, L., Blum, M., and Shub, M. (1986). A simple unpredictable pseudo-random number generator. Advances in Cryptology–CRYPTO.

[9] Boneh, D. and Shoup, V. (2020). Graduate Cryptography.

[10] Brassard, G., Hoyer, P., and Tapp, A. (1997). Quantum cryptanalysis of hash and claw-free functions. Latin American Symposium on Theoretical Informatics.

[11] Bünz, B., et al. (2018). [Titre complet de l'article]. [Nom de la conférence ou revue].

[12] Canetti, R. (2001). Universally composable security: A new paradigm for cryptographic protocols. Proceedings of the 42nd IEEE Symposium on Foundations of Computer Science.

[13] Childs, A. M., Ivanyos, G., and others. (2010). Quantum computation of discrete logarithms in semigroups. arXiv preprint arXiv:1010.1357.

[14] Childs, A. M. (2014). Quantum Information Processing in High-Genus Curves. Quantum Information Computation, 14(11-12), 1081–1096.

[15] De Feo, L., Jao, D., and Plût, J. (2011). Towards quantum-resistant cryptosystems from supersingular elliptic curve isogenies. Post-Quantum Cryptography.

[16] Faugère, J. C. (1999). A new efficient algorithm for computing Gröbner bases without reduction to zero (F5). Journal of Symbolic Computation.

[17] Flajolet, P., and Odlyzko, A. M. (1994). Random mapping statistics. In Advances in Cryptology—EUROCRYPT'89 (pp. 329-354). Springer, Berlin, Heidelberg.

[18] Flajolet, P., and Sedgewick, R. (1990). Analytic combinatorics—functional equations, rational and algebraic functions. Research Report RR-1180, INRIA.

[19] Garey, M. R. and Johnson, D. S. (1979). Computers and intractability: A guide to the theory of NP-completeness. Freeman.

[20] Giovannetti, V., Lloyd, S., and Maccone, L. (2013). Oracles with costs. Advances in Cryptology–CRYPTO.

[21] Goldreich, O., Goldwasser, S., and Micali, S. (1986). How to construct random functions. Journal of the ACM.

[22] Goldreich, O. (2001). The Foundations of Cryptography - Volume 1: Basic Tools. Cambridge University Press.

[23] Grassi, L., Khovratovich, D., Rechberger, C., Roy, A., and Schofnegger, M. (2018). MiMC: Efficient encryption and cryptographic hashing with minimal multiplicative complexity. Advances in Cryptology–ASIACRYPT.

[24] Grassi, L., Onofri, S., Pedicini, M., and Sozzi, L. (2021). Polynomials and their (iterated) reductions. Designs, Codes and Cryptography.

[25] Hansen, T. and CFRG. (2020). Universally Composable Cryptographic Protocols. IRTF Crypto Forum Research Group.

[26] Lidl, R. and Niederreiter, H. (1997). Finite Fields. Cambridge University Press.

[27] Mahmoody, M. and Pass, R. (2018). NP-hardness of equations over finite fields. SIAM Journal on Computing.

[28] Magniez, F., Nayak, A., Roland, J., and Santha, M. (2007). Search via quantum walk. SIAM Journal on Computing.

[29] Minka Mi Nguidjoi, T. E. (2025). The AIIP Problem: Toward a Post-Quantum Hardness Assumption from Affine Iterated Inversion over Finite Fields. Cryptology ePrint Archive.Report 2025/1590.

[30] Minka Mi Nguidjoi, T. E., Mani Onana, F. S., and Djotio Ndié, T. (2025). The CRO trilemma: A formal incompatibility between confidentiality, reliability and legal opposability in post-quantum proof systems. Cryptology ePrint Archive, Report 2025/1348.

[31] Minka Mi Nguidjoi, T. E., Mani Onana, F. S., Djotio Ndié, T., and Bouetou Bouetou, T. (2025). Quantum composable and contextual security infrastructure (Q2CSI): A modular architecture for legally explainable cryptographic signatures. Cryptology ePrint Archive, Report 2025/1380.

[32] Albrecht, M. et al. (2016). MiMC: Efficient Encryption and Cryptographic Hashing with Minimal Multiplicative Complexity. EUROCRYPT.

[33] Moh, T. (1999). The hardness of solving multivariate quadratic equations. Journal of Complexity.

[34] Patarin, J. (1996). Hidden fields equations (HFE) and isomorphisms of polynomials (IP): Two new families of asymmetric algorithms. Advances in Cryptology—EUROCRYPT.

[35] Peikert, C. (2016). A decade of lattice cryptography. Foundations and Trends in Theoretical Computer Science.

[36] Grassi, L. et al. (2021). Poseidon: A New Hash Function. EUROCRYPT.

[37] Aly, A. et al. (2020). Rescue: A Standard of Reliable Encryption. CRYPTO.

[38] Rivest, R. L., Shamir, A., and Wagner, D. A. (1999). Time-lock puzzles and timed-release crypto. MIT LCS TR.

[39] SageMath. (2023). Open-Source Mathematics Software.

[40] Shor, P. W. (1997). Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer. SIAM Review.

[41] Silva, C. E. (2008). Invitation to Ergodic Theory. American Mathematical Society.

[42] Stichtenoth, H. (2009). Algebraic function fields and codes. Springer Science & Business Media.

[43] Unruh, D. (2014). Quantum proofs of knowledge. In Advances in Cryptology-EUROCRYPT 2012 (pp. 135-152). Springer, Berlin, Heidelberg.

[44] Weil, A. (1948). On some exponential sums. Proceedings of the National Academy of Sciences.

[45] Zhandry, M. (2012). How to construct quantum random functions. In 2012 IEEE 53rd Annual Symposium on Foundations of Computer Science (pp. 679-687). IEEE.

[46] Anonymized Directory `https://github.com/KryptoResearcher/CEE`